



General guidelines for the optimal economic aggregation of prosumers in energy communities

Gabriele Volpato^a, Gianluca Carraro^{a,*}, Marco Cont^a, Piero Danieli^a, Sergio Rech^b,
Andrea Lazzaretto^a

^a University of Padova, Industrial Engineering Department, Via Venezia 1, 35131, Padova, Italy

^b Veil Energy Srl, San Pietro in Gu, Padova, Italy

ARTICLE INFO

Keywords:

Citizen energy community
Renewable energy community
MILP optimization
Cost allocation criterion
Demand response

ABSTRACT

Energy communities are regulatory tools promoting aggregations of users to foster the shift towards a renewable distributed generation. First in the literature, this paper addresses together three main aspects affecting the convenience of these aggregations: the complementarity between generation and demand of different prosumers, the criterion allocating the operating costs of energy communities, and the application of demand-response programs. The goal is quantifying the relative weight of these aspects using Mixed-Integer Linear Programming to minimize the operating costs of citizen and renewable energy communities, where prosumers are connected to the grid as single entity, or separately. Incentive- or price-based demand-response programs and a novel cost allocation criterion, which rewards the members with the highest economic benefit in passing from simple consumers to prosumers, are applied to each community configuration. Results allow identifying general guidelines for the optimal economic operation of energy communities: i) complementarity may reduce costs by 15–20%, ii) a fairer cost allocation criterion may reduce the bills of prosumers using free-of-charge renewables by 20–30% compared to those using dispatchable sources, and iii) price-based demand-response may reduce community costs beyond 50%. Eventually, directions of further research, as the impact of energy communities on a national power system, are drawn.

1. Introduction

Several Countries are currently supporting an enhanced use of Renewable Energy Sources (RES) to be aligned with the decarbonization strategies that are required to counteract the climate change. Recently, the “Clean Energy Package” [1] with its energy and environmental targets, as the emission reduction target of 40% in 2030 [2], were updated by the European Union. The new targets, aimed at achieving the carbon neutrality by 2050 [3], refer to a RES share of 40%, an increase in energy efficiency by 36% in the final energy consumption and a reduction of emissions by at least 55% compared to the levels in 1990 [4]. The updated targets are expected to bring higher economic, environmental and social benefits to the European citizens [5].

In this context, the concept of “Energy Community” (EC) has recently raised a growing interest for promoting the aggregation of energy users at the local level [6], facilitating the energy sharing and leading to several benefits as economic savings in the energy bills and increased

exploitation of distributed RES. In Europe the recast of the Internal Electricity Market Directive (IEMD) [7] and of the Renewable Energy Directive (RED II) [8] defined the Citizen and Renewable Energy Community (CEC and REC), respectively, shown in Fig. 1. As highlighted by Ref. [9], both directives define the EC configurations as “*legal entities based on the voluntary participation of residential users, small and medium sized enterprises (SMEs) and local authorities, with the main purpose of providing environmental, economic or social benefits rather than attaining financial profits*”. However, the CEC and REC configurations, according to the European policies, present some differences. For instance, the REC can exploit only RES and satisfy different energy demands (electricity, heating, cooling etc.) while the CEC manages only the electrical demand by using renewables and fossil fuel-based energy sources [10]. Moreover, the REC mainly relies on local energy sources (principle of proximity), contrary to the CEC. Candelise and Ruggieri [11] studied the status of ECs in Italy, highlighting the recent trend of fostering local and small citizen-led projects [12]. Indeed, Italy recently established a common legislation about the development of collective

* Corresponding author.

E-mail addresses: gabriele.volpato.1@phd.unipd.it (G. Volpato), gianluca.carraro@unipd.it (G. Carraro), marco.cont@studenti.unipd.it (M. Cont), piero.danieli@unipd.it (P. Danieli), s.rech@veil-energy.eu (S. Rech), andrea.lazzaretto@unipd.it (A. Lazzaretto).

<https://doi.org/10.1016/j.energy.2022.124800>

Received 7 December 2021; Received in revised form 17 June 2022; Accepted 10 July 2022

Available online 15 July 2022

0360-5442/© 2022 Elsevier Ltd. All rights reserved.

Nomenclature			
<i>Agr</i>	Agricultural	<i>PV</i>	Photovoltaic
<i>b</i>	Electricity bill (€)	<i>REC</i>	Renewable Energy Community
<i>c</i>	Costs (€)	<i>RED II</i>	Recast of the Renewable Energy Directive
<i>CEC</i>	Citizen Energy Community	<i>Res</i>	Residential
<i>DLC</i>	Direct Load Control	<i>RES</i>	Renewable Energy Sources
<i>DR</i>	Demand Response	<i>rev</i>	Economic revenues (€/kWh)
<i>E</i>	Energy (kWh)	<i>SC</i>	Self consumption of the energy community (%)
<i>EC</i>	Energy Community	<i>Ter</i>	Tertiary
<i>IBDR</i>	Incentive-based demand response	<i>TOU</i>	Time of Use
<i>ICE</i>	Internal Combustion Engine		
<i>inc</i>	Economic incentive (€/kWh)	<i>Subscripts</i>	
<i>Ind</i>	Industrial	<i>dem</i>	Demand
<i>IEMD</i>	Internal Electricity Market Directive	<i>el</i>	Electricity
<i>MES</i>	Multi-Energy System	<i>gen</i>	Generation
<i>MILP</i>	Mixed Integer Linear Programming	<i>inter</i>	Intermediate
<i>P</i>	Power (kW)	<i>s</i>	Shared
<i>PBDR</i>	Price-based demand response	<i>self</i>	Self-consumed
		<i>t</i>	Time
		<i>tot</i>	Total

self-consumption and RECs [13]. Lowitzsch et al. [14] explained that the EC represents the “governance model” of a “renewable energy cluster”, featured by the heterogeneity and complementarity of the energy sources and demands, the presence of flexibility options (storage, demand response etc.) and the bi-directionality of energy flows. In this perspective, the main object of this research on ECs is the aggregation of different prosumers within an EC, focusing on how the demand side can be modified to profitably meet the generation one.

1.1. Literature review

ECs can generate several benefits for the overall “society” [15]. Economic gains may result from the optimal installation of photovoltaic (PV) plants and the PV sharing in ECs [16] and the optimal establishment of ECs in city districts [17]. Also technical benefits as the improved reliability of the power supply [18] and the enhanced self-consumption and autonomy [19] can be obtained by ECs. “Net zero” ECs, i.e., fed by renewables and having a null balance of the emissions in the whole life-cycle, can achieve relevant primary energy savings compared to conventional systems exploiting fossil fuels [20]. Feed-in tariff mechanisms that reward the members of RECs for their avoided CO₂ emissions could lead to important reductions of power system emissions [21]. ECs

could also promote the rural electrification planning [22] and a wider access to renewables for vulnerable energy users [23]. Other works dealing with the economic, technical, environmental and social advantages of ECs follow.

Di Lorenzo et al. [24] developed a model for the aggregation of users in buildings, including only one prosumer between the electrical grid and the consumers. In this configuration, the cost savings of each consumer were at least of 36%. Fichera et al. [25] modelled an urban EC including residential buildings with PV systems that are connected by an electric microgrid. The power exchange among the buildings resulted in energy savings from 12 to 99% compared to the configuration without power exchanges among buildings. Ceglia et al. [26] obtained a PV-electricity self-consumption of up to 79% for a REC comprising of two PV plants having different peak power (9 and 14.25 kW), electric heat pumps and one charging station for electric vehicles. Bartolini et al. [27] found that the promotion of self-consumption within an urban EC, avoiding in turn uncontrollable electricity injections into the grid, ensures an enhanced stability. Ugwoke et al. [28] performed a community-scale energy planning of two rural communities in Nigeria. Different scenarios projected to 2040 showed that energy efficiency strategies could reduce the greenhouse gas emissions by 0.3–16% compared to a reference scenario of 2015. Barone et al. [29] found that

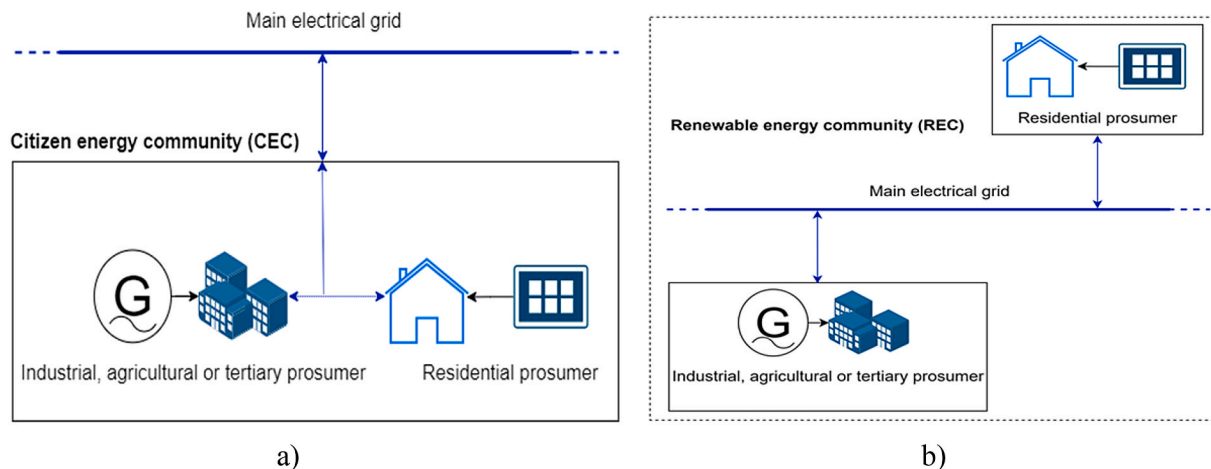


Fig. 1. a) Citizen and b) Renewable energy community including an industrial, agricultural or tertiary prosumer and a residential one equipped with an ICE (G) and PV, respectively.

the integration of a wind-power pumped hydro storage and solar thermal plants may result in 28 kton/year of avoided CO₂ emissions in the island community of El Hierro. Liu et al. [30] carried out a multi-objective design-operation optimization and a sensitivity analysis on carbon tax, electricity price and equipment price, for a nearly-zero EC, achieving a reduction of the annual carbon emissions by 52–73%. Kotagodahetti et al. [31] proposed a multi-criteria approach to evaluate different strategies of emission reduction in community-scale energy systems. Li et al. [32] optimized the sizes of a fleet of PV plants, wind turbines and biogas fueled Diesel generators to ensure the sustainable electrification of an off-grid community in West China. Aviso et al. [33] presented a fuzzy optimization to meet multiple demands of a remote energy community, exploiting micro-hydropower turbines under drought conditions.

A key role in enhancing the benefits of aggregation in ECs may be played by Demand Response (DR) programs [34]. Price-Based DR (PBDR) programs [35] affect the energy demands of customers by setting time-varying electricity tariffs, whereas Incentive-Based DR ones (IBDR) [36] ensure incentives for load reductions in particular periods. The main benefits of DR are the lower energy bills, the improvement of reliability in the power grid and the mitigated environmental impact [37].

Relatively few works investigated the potential of DR programs to enhance the economic convenience of ECs. Geneidy and Howard [38] proposed an IBDR program for a residential community equipped with air-source heat pumps. By preheating the buildings before the DR period, they obtained a demand reduction from 6 to 9 kW while ensuring the thermal comfort of the inhabitants. Qiu et al. [39] exploited the use of interconnected distributed energy sources and PBDR, in detail a Time-of-Use (TOU) program, to achieve a reduction of the annual costs and carbon emissions of 7.3 and 32.1%, respectively, in a Chinese community.

The above-mentioned works evaluate the benefits of being part of an EC and the strategies (e.g., DR programs) that enhance them. However, most of these works do not focus on the principles encouraging the EC aggregation. One of these is the “complementarity” of the prosumers, which helps achieve the optimal coupling among different generation and demand profiles referring to different sectors. A second principle is the definition of a set of rules that guarantee the coalition stability among all members. One of the most important rules is the allocation criterion of the EC costs/revenues. As reported by Berka and Creamer [40] and Gjorgievski et al. [41], this aspect is still not exhaustively addressed in the literature. Li et al. [42] presented different criteria of cost allocation in the design of tariffs for large power systems and they proposed approaches to adapt these criteria in local ECs. However, a specific criterion is suitable for an EC if it is consistent with its configuration and goals. For example, if investors need cost recovery within a renewable-based EC, an allocation criterion relying on the marginal cost of the community (as one of those described in Ref. [42]) is not suitable owing to the low value of the marginal cost.

1.2. Research gap, goal and novelty of the paper

All the above-mentioned aspects, i.e., complementarity, cost allocation criterion and DR strategies, must be considered together to carry out a thorough evaluation of the positive impact of ECs on the current energy system scenario. To the authors' knowledge, only few papers in the current literature dealt with more than one aspect together. Wang et al. [43], Li et al. [44], and Li et al. [45] carried out design, operation, and design-operation optimization procedures, respectively, of communities having different energy conversion units (e.g., PV, wind turbines, micro gas turbines, electric and absorption chillers etc.) and storage systems (electrical, thermal, gas). They also implemented PBDR and IBDR programs. However, in these works the complementarity of the prosumers was not addressed specifically. In fact, all previous works modelled ECs as a single user, i.e., they considered an aggregated

demand (represented by the sum of the individual demands of the users) and neglected the separate behaviours of users and their level of complementarity. Moreover, the authors did not propose any cost allocation criterion. Cosic et al. [46] developed a Mixed-Integer Linear Programming (MILP) model for the optimal planning and operation of an Austrian REC, with different TOU tariffs for the EC members. The investment on new centralized PV plants and electrical energy storage systems reduced the total annual costs of 15.3%. Nevertheless, this model did not allocate the EC costs among its members. Reis et al. [9] developed a multi-agent model of an EC with residential and not-residential loads (e.g., office, restaurant and school), TOU tariffs, and a cost distribution among members according to their contributions to the overall net power demand. A community self-consumption equal to 89% was calculated in one typical summer week. However, other DR programs (e.g., IBDR) were not evaluated and only residential prosumers equipped with PV were modelled, therefore neglecting the potential complementarity among different types of prosumers. Jiang et al. [47] presented an incentive based mechanism to foster the optimal cooperation between prosumers equipped with PV and the energy community manager, without taking into account any DR program and without considering different types of prosumers.

It clearly emerges that none of the most recent papers in the literature addressed together the fundamental aspects of complementarity, cost allocation and DR programs to quantify their relative weight on the benefits of the EC formation and operation. Conversely, to fill this research gap, the novelty of this paper consists in analysing all these aspects together by an optimization procedure with the aim of evaluating their positive impact on the economic convenience of ECs. The goal is to identify general guidelines, in economic terms, for the optimal aggregation (and subsequent operation) of prosumers having heterogeneous energy conversion units and demand profiles. This is achieved by:

- i. Optimizing the coupling of energy demand and generation profiles of the whole community by taking advantage of the complementarity between those of the individual prosumers. The demand profiles of the single prosumers are chosen to be representative of different sectors (industrial, agricultural, tertiary and residential);
- ii. Developing a fair and novel criterion for the allocation of EC costs among its members;
- iii. Implementing both PBDR and IBDR programs to modify smartly the demands of the different prosumers. This is done here assuming possible variations of the hourly energy demand within ranges of 10% for the PBDR and 10% or 20% for the IBDR, while keeping the daily energy demand unvaried. The higher range for the IBDR is justified by the higher number of energy users usually prone to participate in this kind of program.

For each of the above aspects, another novelty of this work is to consider two different configurations of ECs, in accordance with the European directives defining the CEC and REC configurations. These are optimized and compared, also by taking as an example the Italian legislation framework that includes the contribution of government incentives for the REC.

2. Methods

This Section presents the mathematical models of the two EC configurations considered in this work (i.e., CEC and REC). Section 2.1 reports the input data (demand profiles and type, size and number of the conversion units) and outlines the common structure of the models, whereas Section 2.2 describes in detail each configuration and highlights the main modelling differences.

2.1. Modelling

The models of the EC configurations are built according to the Mixed-Integer Linear Programming (MILP) approach [48]. They are used in an operation optimization procedure that searches for the minimum operating costs of the community. The optimization is solved by means of the Gurobi software [49] with implementation of the linear-programming based branch-and-bound algorithm, which guarantees the global optimality of the solutions.

The MILP optimization problem is formulated as in Eq. (1) [50]:

$$\begin{aligned} \min_{x,y} & (c^T x + d^T y) \\ \text{s.t. } & Ax + By = b \end{aligned} \quad (1)$$

where $x \geq 0 \in \mathbb{R}^{N_x}$, $y \in \{0, 1\}^{N_y}$

in which c and d are the cost arrays associated with the continuous and binary variables, x and y , respectively; A and B are the constraint matrices and b is the known term; N_x and N_y indicate the dimension of x and y , respectively.

This general formulation is specified for the case of the ECs studied in this work by considering the following assumptions and boundary conditions:

- The sizes of the conversion units are given as input and are not part of the decision variables set. Thus, the optimization problem evaluates only the EC operation. The presence of binary variables in the set of

decision variables allows, on one side, to optimize the hourly on/off status of the given units and, on the other side, to differentiate the price ranges in the price-based DR program (Section 3.3.1).

- To understand better the logic behind the aggregation of prosumers in an EC and to keep an acceptable computational time of the optimization procedure, only the coupling of two prosumers at a time is evaluated. However, it is worth observing that each of the two prosumers can represent a group of several prosumers, each one having conversion units with lower sizes. For example, a prosumer with a photovoltaic (PV) plant of 200 kW of peak power represents 20 residential prosumers, each having an installed PV capacity of 10 kW (maximum value of PV peak power at residential level [51]).
- For simplicity, and without loss of generality, one dispatchable energy conversion unit (Internal Combustion Engine, ICE) and one not-dispatchable unit (PV system) are considered when coupling the prosumers of both CEC and REC ECs.
- Four different demand profiles are considered, as representative of the industrial (Ind), agricultural (Agr), tertiary (Ter) and residential (Res) sectors, respectively, and are given as input to the MILP model. In particular, Fig. 2 (a)-(b)-(c)-(d) shows the representative demand of a prosumer in the industrial, agricultural, tertiary and residential sector, respectively, which were taken from Ref. [52].
- According to the European directives that define the ECs, only the generation and demand of electricity is considered to allow a fair comparison between the CEC and REC configurations.
- DR programs (Section 3.3) are a useful strategy to increase the operation flexibility and are applied to shiftable loads. The use of electric storage is not considered since the perspective of the work is

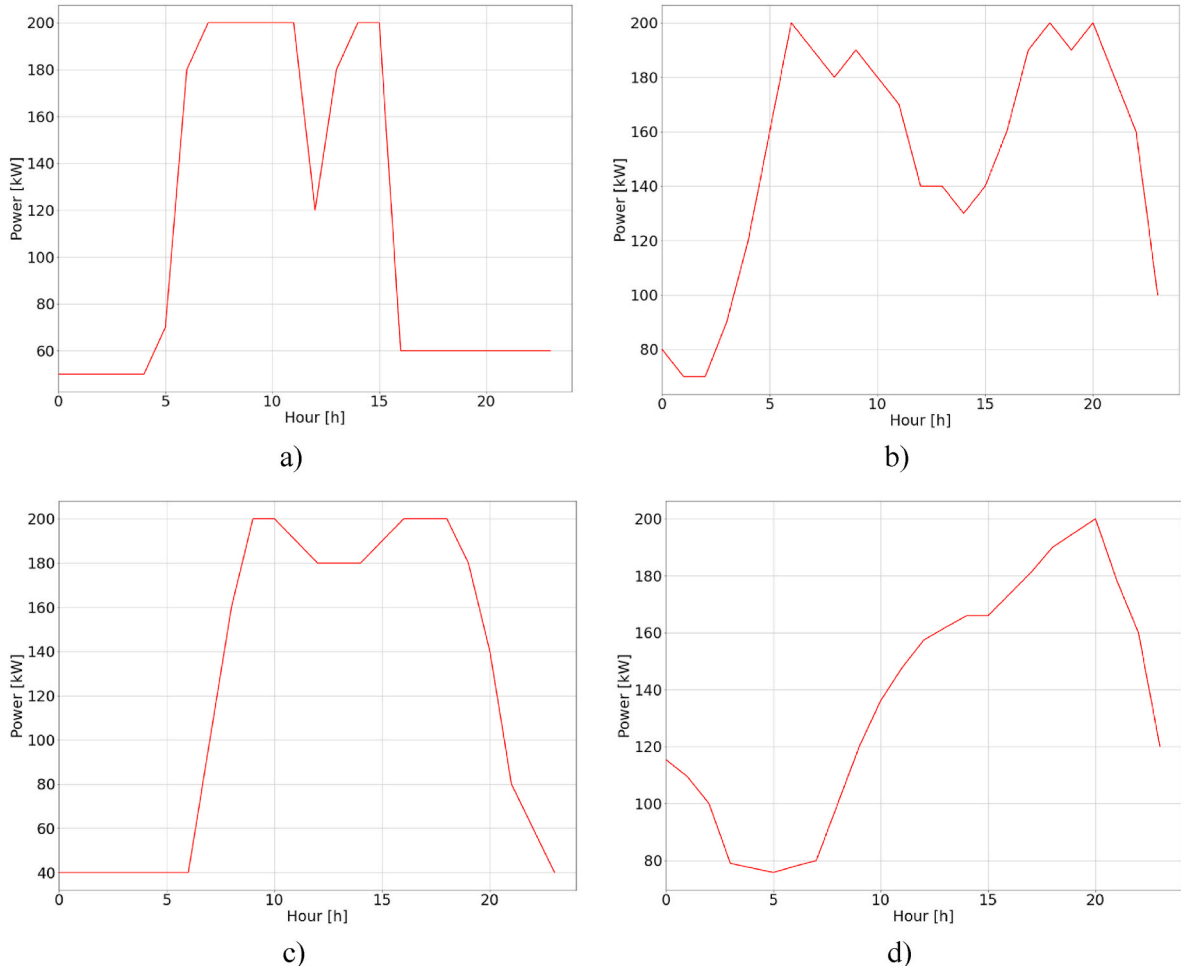


Fig. 2. Daily profiles of the electrical demands of four different sectors taken from Ref. [52]: a) industrial, b) agricultural, c) tertiary, and d) residential.

that of pushing towards the optimal match between generation and demand (i.e., the optimal aggregation between prosumers) limiting to the maximum the need of storage (which is an extra cost for the community).

- The operation optimization of both CEC and REC is carried out for each characteristic day of April, December, and August, as representative of the middle, winter, and summer season, respectively, considering the solar irradiance for the analysed location (Padua, in Italy). Yearly averaged profiles of the electrical demands are taken from the literature due to the lack of more specific available data. However, it should be noted that the variability of the electrical demands across seasons is already considered by analysing the combinations of these demands, characterized by different shapes, which are representative of different prosumers. Hence, the evaluation of a set of several seasonal demands for each type of prosumer would not lead to different economic outcomes compared to those reported in Sections 4.1, 4.2 and 4.3, but it would significantly result in higher computational times.

The objective function in Eq. (1) assumes a different form for the CEC and REC EC configurations, while including the same terms: the costs of fuel, the costs of the energy purchased from the grid, the revenues from the energy sold to the grid and the revenues from any incentives. The detailed equation of the objective function of each EC configuration is shown in Section 2.2.

The ICE is fueled by biogas and its operation is modelled with the following experimental equations [48]:

$$F_{ICE,t} = a_0 \cdot \delta_{ICE,t} + a_1 \cdot P_{ICE,t} \quad (2)$$

$$P_{ICE,t} \leq P_{ICE,max} \cdot \delta_{ICE,t} \quad (3)$$

$$P_{ICE,t} \geq P_{ICE,min} \cdot \delta_{ICE,t} \quad (4)$$

where $F_{ICE,t}$ [kW] and $P_{ICE,t}$ [kW] are the power consumed and produced by the ICE at the time step t , respectively, while $\delta_{ICE,t}$ [-] is the binary variable describing the on/off status of the ICE. Eq. (2) represents the linear form of the input-output relationship for an energy conversion unit, as the ICE, having one fuel stream and one product stream. The values of the parameters a_0 and a_1 (equal to 91 [kW] and 2.008 [-], respectively) are obtained from the data of the manufacturer reported in Ref. [53]. Constraints (3) and (4) are required to fix the upper and lower bounds of $P_{ICE,t}$, respectively. As reported in Ref. [48], approximating the operating behaviour of ICE with this linear equation introduces a small loss of accuracy that is well compensated by the reduction in the computational time to solve the MILP optimization problem.

The operation of the PV plant is described by Eq. (5):

$$P_{PV,t} = \eta_{PV} \cdot A_{PV} \cdot I_{sol,t} \quad (5)$$

where $P_{PV,t}$ [kW] is the hourly power generated by the PV plant, η_{PV} [%] is the average efficiency of the PV plant, A_{PV} [m²] is the area of the PV panels and $I_{sol,t}$ [kW/m²] is the hourly solar irradiance (derived from PVGIS database [54]) on a surface in Padua tilted of 30° to the South. Table 1 lists the input data used for both the CEC and REC optimization models.

Table 1
Input parameters of the ECs models.

Parameter	Value
A_{PV} [m ²]	2546
$P_{ICE,max}$ [kW]	200
$P_{ICE,min}$ [kW]	50
c_{biogas} [€/kWh]	0.055
η_{PV} [%]	0.11

2.2. Configurations of energy communities

The CEC and REC models are analysed in Sections 2.2.1 and 2.2.2, respectively. In this paper, the self-consumption of the EC (CEC and REC) is assessed by the energy self-consumption (SC) ratio, defined as ratio of the total energy self-consumed and total energy consumption (Eq. (6)):

$$SC = \frac{\sum_{t=0}^{23} E_{self,t}}{\sum_{t=0}^{23} E_{el,t}} \cdot 100 \% \text{ for } t \in (0, \dots, 23) \quad (6)$$

where each term $E_{self,t}$ is the minimum between the hourly energy consumption (i.e., the sum of the consumptions of the prosumers, $E_{el,t}$) and generation.

2.2.1. Prosumers connected as a whole entity to the distribution grid

Fig. 1(a) shows the CEC configuration, according to the recast of the IEMD [7,10], where the prosumers are assumed to be the owners of a microgrid. Hence, the CEC constitutes a single entity, from the viewpoint of the distribution system operator, and exchanges electricity with the main electrical grid in one single point. This EC configuration allows internal power exchanges between prosumers which contribute to the “physical self-consumption” within the microgrid of the CEC.

The energy balance (satisfied in each hour t) is formulated as in Eq. (7):

$$E_{ICE,t} + E_{PV,t} + E_{PURCHASE,GRID,t} - E_{SELL,GRID,t} = E_{el,t} \text{ for } t \in (0, \dots, 23) \quad (7)$$

where $E_{ICE,t}$, $E_{PV,t}$, $E_{PURCHASE,GRID,t}$ and $E_{SELL,GRID,t}$ are the energy [kWh] generated by the ICE, generated by the PV, purchased/sold from/to the grid.

Moreover, the internal power exchange between prosumers within the CEC is free of charge and it does not affect the total costs of the EC, described by Eq. (8):

$$c_{CEC} = c_{ICE} + c_{PURCHASE,GRID,CEC} - c_{SELL,GRID,CEC} \quad (8)$$

where c_{ICE} [€], $c_{PURCHASE,GRID,CEC}$ [€] and $c_{SELL,GRID,CEC}$ [€] are the ICE cost for fuel consumption ($F_{ICE,t}$, Eq. (2)), cost for the energy purchased from the grid and revenues for the energy sold to the grid, respectively. They can be expressed as:

$$c_{ICE} = \sum_{t=0}^{23} F_{ICE,t} \cdot c_{biogas} \cdot 1 \text{ h} \quad (9)$$

$$c_{PURCHASE,GRID,CEC} = \sum_{t=0}^{23} E_{PURCHASE,GRID,t} \cdot c_{PURCHASE} \quad (10)$$

$$c_{SELL,GRID,CEC} = \sum_{t=0}^{23} E_{SELL,GRID,t} \cdot c_{SELL} \quad (11)$$

where $c_{PURCHASE}$ is assumed to be the mean electricity price for customers in Italy [55] (0.19 €/kWh) and c_{SELL} is the hourly selling price in Italy [56], shown in Fig. 3.

2.2.2. Each prosumer connected separately to the distribution grid

Fig. 1(b) shows the REC configuration according to the Italian legislation [13]. All prosumers are directly and separately connected to the same (main) low-voltage distribution grid, which encompasses the power exchanges between prosumers contributing to the “virtual self-consumption” of the REC. The energy balances describe separately the total energy withdrawn from ($E_{withdrawn,t}$) and injected to ($E_{injected,t}$) the grid as reported by Eq. (12) and Eq. (13), respectively:

$$E_{withdrawn,t} = E_{dem,t} = E_{el,t} + E_{c,sto,t} \text{ for } t \in (0, \dots, 23) \quad (12)$$

$$E_{injected,t} = E_{gen,t} = E_{ICE,t} + E_{PV,t} \text{ for } t \in (0, \dots, 23) \quad (13)$$

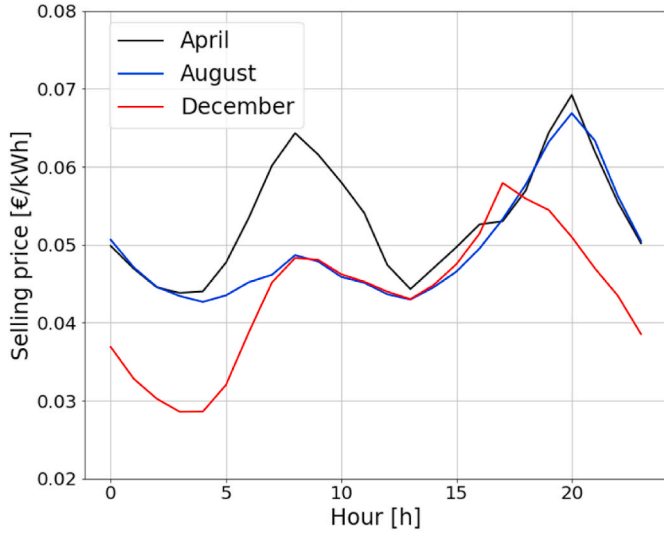


Fig. 3. Daily profile of selling price for a characteristic day of April, August and December in Italy.

Eq. (12) and Eq. (13) represent the total energy demand ($E_{dem,t}$) and the total energy generated ($E_{gen,t}$), respectively. In this case study, the total energy demand is equivalent to the total energy consumed ($E_{el,t}$) because there is not the contribution of energy charged into a storage unit ($E_{c,sto,t}$). From this point on, the energy demand is meant as energy consumed.

The costs of the energy exchanged with the grid are now dependent on the definitions of $E_{withdrawn,t}$ and $E_{injected,t}$, as shown by Eqs. (14) and (15):

$$C_{PURCHASE,GRID,REC} = \sum_{t=0}^{23} E_{withdrawn,t} \cdot C_{PURCHASE} \quad (14)$$

$$C_{SELL,GRID,REC} = \sum_{t=0}^{23} E_{injected,t} \cdot C_{SELL} \quad (15)$$

According to the Italian legislation, the REC benefits from an incentive for the “shared energy” among its members, which is defined as the minimum between $E_{withdrawn,t}$ and $E_{injected,t}$:

$$E_{s,t} = \min(E_{withdrawn,t}, E_{injected,t}) \text{ for } t \in (0, \dots, 23) \quad (16)$$

The incentive (inc_{REC} in Eq. (17)) is fixed to 0.12 €/kWh (similarly to the Italian value), which is the sum of a feed-in premium (0.11 €/kWh) and a feed-in tariff (around 0.008–0.01 €/kWh) linked to the avoided network losses within the REC.

The total costs of REC are:

$$C_{REC} = C_{ICE} + C_{PURCHASE,GRID,REC} - C_{SELL,GRID,REC} - inc_{REC} \cdot \sum_{t=0}^{23} E_{s,t} \quad (17)$$

where the last term represents the economic revenues due to the incentive.

3. Aggregation of prosumers

This Section identifies and analyses the aspects that foster the aggregation of prosumers in ECs. These aspects refer to the complementarity between energy demand and generation profiles of different prosumers (Section 3.1), the implementation of a fair cost allocation criterion (Section 3.2) and of DR programs (Section 3.3).

3.1. Complementarity between prosumers

The aggregation between prosumers results in economic benefits compared to the case of their separate operation. Indeed, the power generated by one prosumer can help meet the demand of a second prosumer in case the latter faces high generation costs or is not provided with generation sources and storage units in the considered time framework (e.g., if the second prosumer is equipped with PV only, it cannot use PV power during the night).

The most economically convenient aggregation between two prosumers occurs when the conversion units are complementary in relation to the respective electrical demands. For example, consider the normalized electrical demand profiles reported in Fig. 4. Assume that the prosumers 1 and 2 have the blue and green demands with a dispatchable conversion unit (e.g., ICE with a size lower than the demand peak) and a PV plant, respectively. In the case of separate operation, the ICE of prosumer 1 cannot fulfil the peak load of its demand whereas the PV of prosumer 2 cannot meet the constant load required in the morning and in the evening. On the contrary, in the case of a joint operation, the PV of prosumer 2 could meet the demand peak of prosumer 1 and the ICE of prosumer 1 could satisfy the demand of prosumer 2 in absence of solar power. Thus, the complementarity of the two prosumers and the coordinated operation of the conversion units is a cost-effective solution with also an improved efficiency for the ICE (operating at constant load for most of the day instead of few hours in the middle of the day).

3.2. Cost allocation criterion

One of the main issues of ECs is the definition of a fair criterion to allocate costs/revenues among all the members guaranteeing the energy “distributive justice” [57]. Fig. 5 reports the economic flows within the governance model of an EC where the aggregator, assumed here to be a non-profit agent, manages all the incomes and expenditures of the EC. Fig. 5 shows how the bills of prosumers before allocation (block 1) are used by the aggregator to afford the total operating costs of the community (block 2). Subsequently, the remaining revenues after subtracting costs (block 3) are distributed among the prosumers according to a cost allocation criterion (block 4).

Different cost allocation criteria could be implemented in ECs. The criterion proposed here suggests that the bill of each prosumer p belonging to the EC ($b_{p,new}$) is calculated according to the following formula:

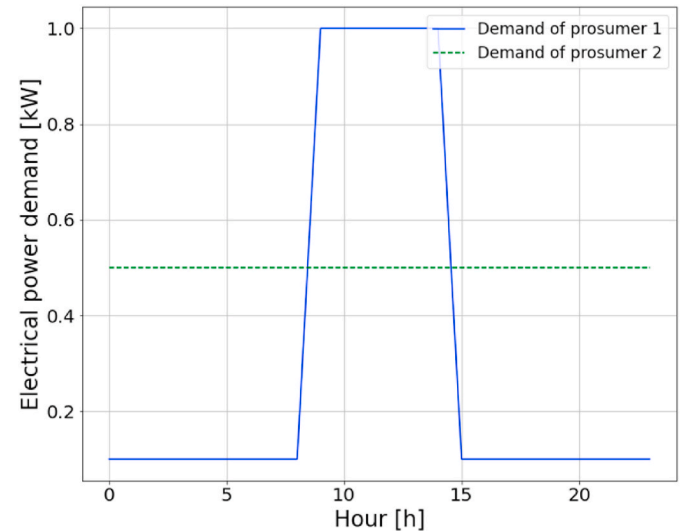


Fig. 4. Normalized demand profile with a peak load in the middle of the day (blue) and constant demand (green) for two prosumers.

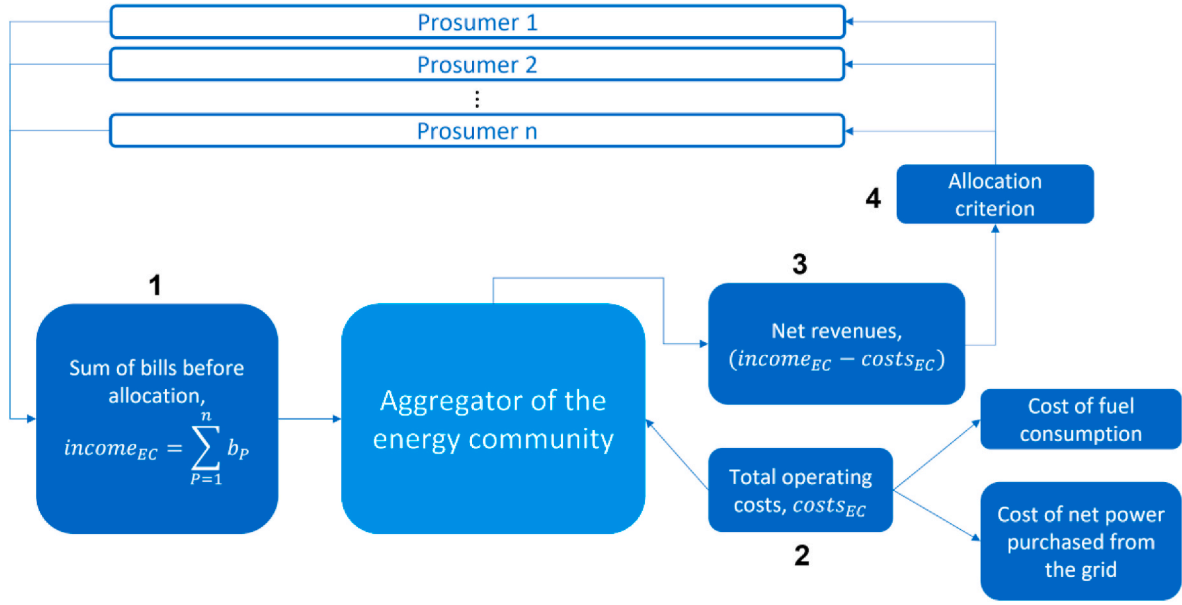


Fig. 5. Economic flows managed by the aggregator of the energy community.

$$b_{p,new} = b_p - (income_{EC} - costs_{EC}) \cdot \frac{(b_{p,0} - b_{p,1})}{\sum_{p=1}^n (b_{p,0} - b_{p,1})} \quad (18)$$

where:

- b_p is the bill of the prosumer p within the EC before allocation;
- $income_{EC}$ is the sum of all the electricity bills (b_p) before allocation;
- $costs_{EC}$ are the total costs (i.e., the objective function to minimize, Eq. (8) or Eq. (17));
- $(b_{p,0} - b_{p,1})$ is the economic saving of the prosumer p achieved by moving from the status of pure consumer (having the total electrical demand fulfilled by the grid and the bill $b_{p,0}$) to the status of independent prosumer, i.e., not belonging to the EC (having the bill $b_{p,1}$ lower than $b_{p,0}$);
- n is the number of prosumers.

Eq. (18) subtracts from the bill b_p (before the distribution of revenues) a part of the net revenues ($income_{EC} - costs_{EC}$) of the EC, proportionally to the economic savings achieved in changing the status from pure consumer to that of independent prosumer. In other words, this criterion encourages the members who have the higher advantage in becoming independent prosumers from the status of consumers, to become part of the EC. The idea is to reward more who finds the better compromise between maximum self-generated energy and minimum cost for self-generation. Thus, it is expected that the renewable generation from PV is best rewarded in most cases (particularly when the solar irradiance is high) leaving in general lower rewards to the prosumer with the ICE, except in the cases of very low irradiance and low cost for the fuel.

Moreover, b_p is the product between the electrical demand of prosumer p ($E_{el,t,p}$) and the purchase price as described by Eq. (19):

$$b_p = \sum_{t=0}^{23} E_{el,t,p} \cdot C_{PURCHASE} \quad (19)$$

which resembles Eq. (14) for the REC. It is worth highlighting that the implementation of a DR program (Section 3.3) affects the bills of prosumers before allocation ($E_{el,t,p}$ in Eq. (19)) and, in turn, the cost allocation criterion (Eq. (18)).

Alternatively, a criterion allocating costs proportionally to the demands of the single prosumers, based on an average price of

consumption [42], could disadvantage the users with highest energy demands. The average price of consumption is calculated as the ratio between the total operating costs of the EC and its total electrical demand. Such a criterion is also applied subsequently to have a comparison with the primary allocation criterion proposed in this paper (Eq. (18)).

3.3. Demand response strategies

This Section describes two DR approaches included in the optimization models of the ECs. The first approach is based on time-varying electricity tariffs (price-based DR, Section 3.3.1), whereas the second one on incentives (incentive-based DR, Section 3.3.2). A detailed overview of the main DR equations and constraints follows for each of the two approaches.

3.3.1. Price-based demand response

Price-Based DR (PBDR) strategies rely on programs with time-varying tariffs. For example, Time-of-Use (TOU) establishes two or three daily tariffs. Real Time Pricing consists of different hourly tariffs following the day-ahead market price and Critical Peak Pricing sets high prices in few hours of the year to prevent the occurrence of high demands during extreme overloads in the power grid.

The PBDR implemented here exploits a TOU program, the most socially accepted from the majority of customers [58], with three electricity tariffs (base, intermediate and peak).

The base and peak electricity prices are expressed by:

$$c_{base} = c_{inter} \cdot \left(1 + \frac{d_L}{El}\right) \quad (20)$$

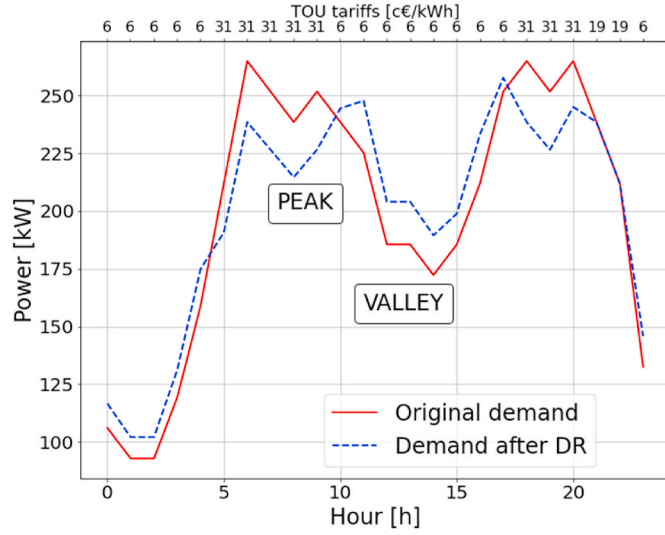
$$c_{peak} = c_{inter} \cdot \left(1 - \frac{d_L}{El}\right) \quad (21)$$

and are equal here to 0.06 €/kWh and 0.31 €/kWh, respectively, where the intermediate price c_{inter} is equal to $c_{PURCHASE}$ (0.19 €/kWh, see Section 2.2.1), d_L is the maximum load change due to DR in a specific hour [59] and El is the price elasticity [60,61] (Table 2). Fig. 6 shows an example of how the TOU program could affect the behaviour of a prosumer with an agricultural demand. Partial load shifts result from periods with the peak price of 0.31 €/kWh (hours 5–9 and 18–20) to periods with the base price of 0.06 €/kWh (hours 0–4, 10–17 and 23)

Table 2

Parameters associated with the PBDR based on TOU tariffs.

Parameter	Value	Source
c_{base} [€/kWh]	0.06	–
c_{peak} [€/kWh]	0.31	–
c_{inter} [€/kWh]	0.19	[55]
d_L	0.1	[59]
El	−0.15	[61]

**Fig. 6.** Peak and base electricity TOU prices could result in a decrease and increase, respectively, of the demand load after the PBDR (blue dashed line) compared to the original one (red line).

[62], whereas the intermediate price of 0.19 €/kWh is found when there is not a noticeable load change after DR (hours 21–22). Hence, the PBDR programs as TOU allow reducing the demand spikes (i.e., peak shaving) and increasing the demand in periods with low load (i.e., valley filling), thus resulting in an improved reliability of the power grid.

The main DR constraints [63], applied for each prosumer, are:

$$\begin{aligned} P_{el,t,DR} &\leq P_{el,t} \cdot (1 + d_L) \text{ for } t \in (0, \dots, 23) \\ P_{el,t,DR} &\geq P_{el,t} \cdot (1 - d_L) \text{ for } t \in (0, \dots, 23) \end{aligned} \quad (22)$$

$$\sum_{t=0}^{23} E_{el,t,DR} = \sum_{t=0}^{23} E_{el,t} \quad (23)$$

$$\begin{aligned} P_{el,t,DR} &\leq \max(P_{el}) \text{ for } t \in (0, \dots, 23) \\ P_{el,t,DR} &\geq \min(P_{el}) \text{ for } t \in (0, \dots, 23) \end{aligned} \quad (24)$$

where $P_{el,t}$ and $P_{el,t,DR}$ are the hourly electrical power demand before and after the application of DR, respectively. Constraints (22) define the range of variation of $P_{el,t,DR}$ in each hour. Eq. (23) states that the daily energy demand after DR (sum of $E_{el,t,DR}$) is unchanged compared to that before DR (sum of $E_{el,t}$), which usually helps promote a high participation of customers to the DR program. Constraints (24) avoid the presence of greater demand spikes after the DR compared to those before DR, therefore ensuring the stability of the main electrical distribution grid. Constraints (24) are applied also for the total electrical demand.

3.3.2. Incentive-based demand response

Incentive-Based Demand Response (IBDR) programs could be voluntary (e.g., Direct Load Control and Emergency DR Program), mandatory (e.g., Interruptible Service) or market-related (e.g., Demand Bidding) [64]. In mandatory programs, contrary to the voluntary ones, customers are penalized if they do not adhere to the load modification.

In market-related programs, customers can offer load reductions to the system operator at a profitable price. Generally, customers accepting these programs are awarded incentives when reducing, shifting or interrupting their loads.

In this paper a Direct Load Control (DLC) program with a unitary incentive, named inc_{IBDR} (Eq. (26)) and equal to 0.08 €/kWh [65], is considered. This incentive is given to a prosumer only when it decreases the hourly power demand by at least 5%:

$$\frac{(P_{el,t} - P_{el,t,DR})}{P_{el,t}} \geq 0.05 \text{ for } t \in (0, \dots, 23) \quad (25)$$

and the daily revenues of a prosumer owing to the IBDR is given by Eq. (26):

$$rev_{IBDR} = \left(\sum_{t=0}^{23} E_{el,t} - E_{el,t,DR} \right) \cdot inc_{IBDR} \quad (26)$$

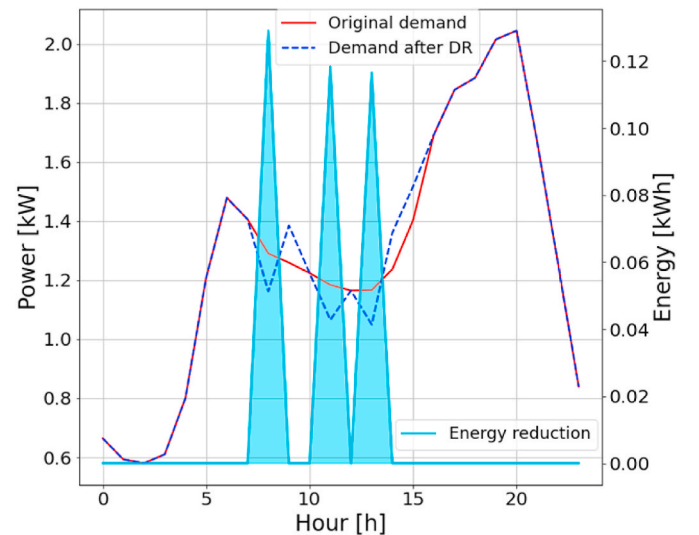
where $(E_{el,t} - E_{el,t,DR})$ is calculated only when Eq. (25) is satisfied.

Constraints (22)–(24) for the PBDR (Section 3.3.1) are still implemented here for the IBDR. Fig. 7 shows an example of application of the IBDR program for a prosumer with a residential demand, and it highlights in light blue the energy reduction achieved in 3 hours of the day (namely, hours 8, 11 and 13). In this example, the presence of the IBDR fosters the prosumer to properly modify its electrical demand to receive the economic benefit related to the reduction of the load.

Anyway, note that the single participants joining this IBDR program are charged a fixed electricity price at all the hours (0.19 €/kWh, Table 2), contrary to the PBDR case. Moreover, the maximum load change in a specific hour (d_L , Table 2), which usually represents the percentage of customers participating to the DR program, is assumed to be equal or higher in the IBDR (i.e., 10 or 20%) compared to that fixed for the PBDR (i.e., 10%). Indeed, customers usually prefer a voluntary DLC than a predefined TOU program.

4. Optimization results

The optimization results of CEC, REC and final critical remarks are discussed in Sections 4.1, 4.2 and 4.3, respectively. The complementarity of prosumers is investigated considering three different couplings, namely the residential prosumer (equipped with PV) with i) the industrial one (Ind-Res), with ii) the agricultural one (Agr-Res) and with iii) the tertiary (Ter-Res) one (equipped with ICE). The total minimized

**Fig. 7.** Example of energy reduction in 3 hours (8, 11, 13) due to the IBDR with DLC.

community costs and the bills of the prosumers, the latter calculated according to the allocation criterion proposed in this paper (shortened as “C1”) and then considering that proportional to the energy demands (shortened as “C2”), are reported for all couplings. Both PBDR and IBDR programs are implemented in the optimization procedure. Sections 4.1 and 4.2 report the outcomes from the optimization carried out on the characteristic day of April, whereas the results referring to December and August are summarized in Section 4.3 and reported in the Appendix. Fig. 8 highlights the different prosumer couplings, the cost allocation criteria and the DR programs considered in the MILP optimization of the CEC and REC EC configurations.

4.1. Citizen energy community

The results discussed in this Section refer to the characteristic day of April. Fig. 9 reports the total community costs for the CEC (Section 2.2.1) and the bills of the prosumers for all the couplings (i.e., Ind-Res, Agr-Res and Ter-Res) in the basic aggregation without DR, with PBDR (Section 3.3.1), with IBDR (Section 3.3.2) and a maximum load change of 10% (named IBDR1) or 20% (named IBDR2). Table 3 displays the SC values of the CEC without and with DR for all the couplings, and the economic savings in the total community costs, compared to the basic aggregation, if a DR program is implemented.

The highest level of complementarity between prosumers is found with the coupling Ind-Res, which achieves the lowest costs (Fig. 9) and the highest self-consumption (SC) of the community (Table 3). In fact, in the basic aggregation, the total community costs (638.42 €) and the SC (94.07%) of Ind-Res are lower and higher than those of Agr-Res (807.68 € and 82.86%) and Ter-Res (713.84 € and 85.12%), respectively. As shown by the separate operation of the industrial and residential prosumers in Fig. 10, the former can exploit the power dispatchability of the ICE to help the latter cover its demand during the morning and late afternoon, whereas the latter can satisfy the demand of the former partially in the middle hours. Thus, the collaborative operation of the industrial and residential prosumers is economically beneficial and highly recommended for the CEC configuration.

In each coupling, the specific criterion used to allocate costs (C1 or C2 in Fig. 9) affects the bills of both prosumers but not the total community costs. The proposed cost allocation criterion C1 results to distribute payoffs that lead to lower bills for prosumers exploiting free-of-charge renewable sources (e.g., solar) compared to those using dispatchable sources (e.g., biogas). In detail, C1 always attains lower economic bills for the residential prosumer, compared to the other ones, because of the higher economic benefit obtained by changing its status from simple consumer to independent prosumer that exploits a free-of-charge renewable source. Consequently, this criterion proves to be favourable (and fair) towards residential prosumers with PV, who cannot exploit the dispatchability of their generation plants, and it encourages them to take part in the EC. Conversely, the application of C2 penalises the prosumer with the highest daily energy demand, e.g., the residential one in Ind-Res and Ter-Res and the agricultural one in Agr-Res.

Since the coupling Ind-Res represents the most convenient aggregation, the optimal power profiles of demand and generation are shown only for this case. Fig. 11 shows the total power demand, generated and exchanged with the grid (i.e., purchased and sold) for the CEC in the different operating conditions, i.e., the basic aggregation without DR (Fig. 11(a)), with the PBDR (Fig. 11(b)), IBDR1 (Fig. 11(c)) and IBDR2 (Fig. 11(d)). Fig. 11 highlights that the power sold to the grid is almost zero in all the operating conditions, meaning that the CEC exploits at maximum the energy SC. Moreover, the DR programs enhance the SC, which increases by about 2% for PBDR and IBDR1, and by 6% for IBDR2, with respect to the basic aggregation. These programs reduce the net energy purchased from the grid and increase the PV utilization in the middle of the day (around hour 13) where the total power demand is increased. Finally, Fig. 11(d) shows that the IBDR2 represents the operating condition with the most constant generation profile of the ICE, thus guaranteeing its optimal operation at maximum efficiency, and with the minimum daily energy purchased from the grid, linked to the highest SC value of 99.68% (Table 3).

It is worth highlighting that the DR programs affect the demand profiles of the single EC members. For instance, Fig. 12 reports the

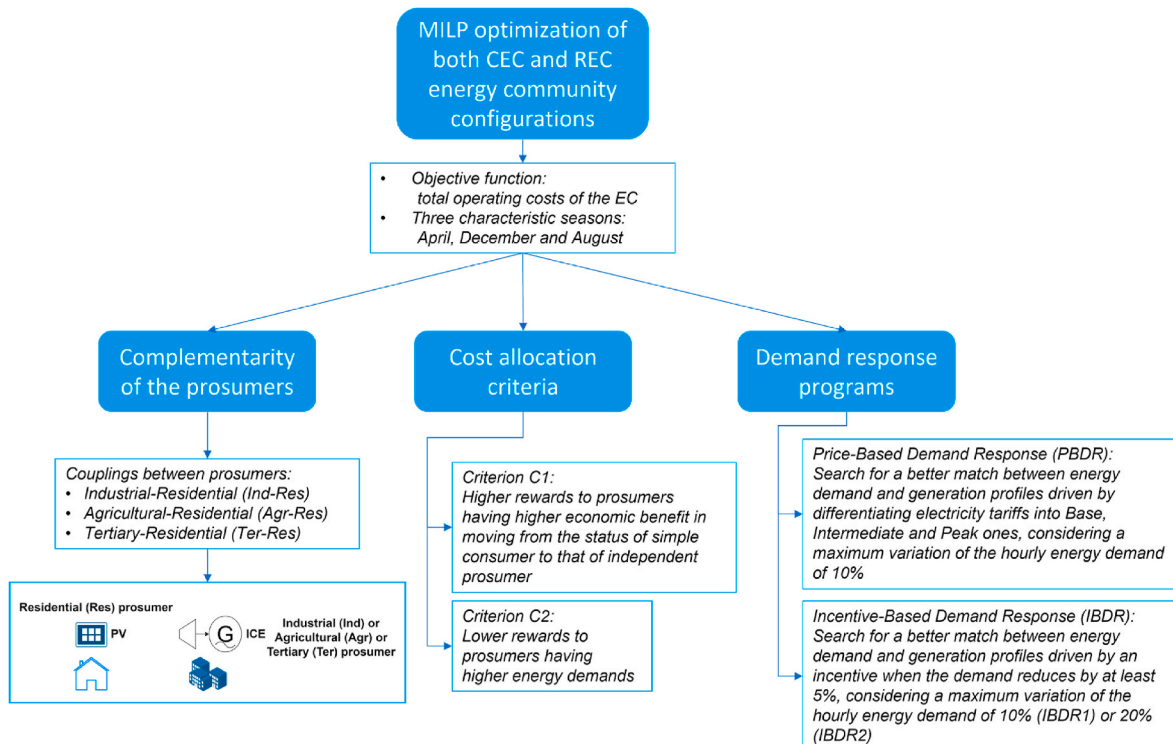


Fig. 8. Flowchart of the optimization procedure highlighting the three aspects that affect the aggregation of prosumers in ECs.

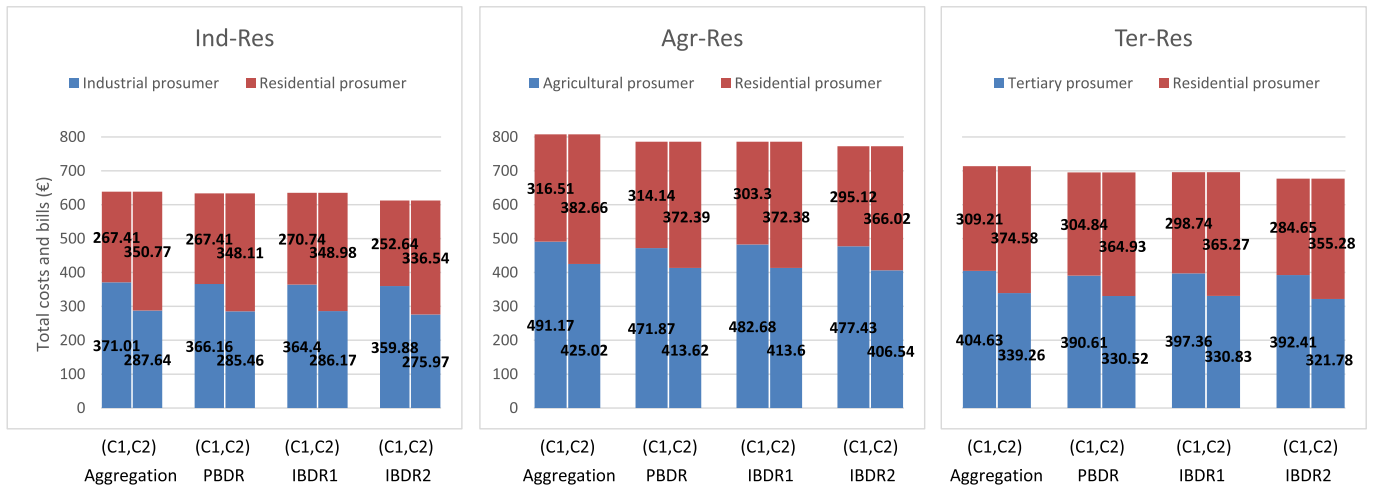


Fig. 9. Economic results from the CEC optimization in April. IBDR1 and IBDR2 configurations refer to a maximum load change of 10% and 20% in a specific hour, respectively.

Table 3

Self-consumption and total economic savings, attained by the application of the DR programs compared to the basic aggregation, in the CEC.

	SC (%)			Savings compared to basic aggregation (%)		
	Ind-Res	Agr-Res	Ter-Res	Ind-Res	Agr-Res	Ter-Res
Basic aggregation	94.07	82.86	85.12	–	–	–
PBDR	95.61	86.91	88.93	−0.76	−2.68	−2.58
IBDR1	96.11	86.91	88.8	−0.51	−2.69	−2.49
IBDR2	99.68	89.42	92.74	−4.06	−4.35	−5.15

power demand before (continuous line) and after (dashed line) the PBDR application, with the TOU tariffs assigned by the optimization, for both the industrial and residential prosumers. It is found that the PBDR leads to a general increase and reduction of the load, for both the residential and industrial prosumers, in the morning and in the evening, respectively, in correspondence of the lowest and highest TOU tariffs equal to 0.06 €/kWh and 0.31 €/kWh. Moreover, Fig. 12 shows that the application of the PBDR program results in a rise of the electrical demand of both prosumers at hour 13, thus explaining the increase of the total demand of the CEC at the same hour as reported in Fig. 11 (b).

The SC increase shown in Table 3 due to the DR strategies entails an

increase of the economic savings for the overall community. For example, the total costs (Fig. 9) with the IBDR1 program in the Agr-Res coupling (SC equal to 86.91%) are 2.69% lower (Table 3) than those in the basic aggregation (SC equal to 82.86%). In all the couplings, the IBDR2 results to be the best DR program in economic (total costs and bills, Fig. 9) and self-consumption (SC, Table 3) terms, owing to the higher share of power demand that can be modified by the program (i.e., 20%), whereas the PBDR and the IBDR1 show similar outcomes. Considering the Ter-Res coupling, the IBDR2 program leads to a SC of 92.74% and to total economic savings of 5.15% (with respect to the basic aggregation), compared to the values of almost 89% and 2.5%, respectively, obtained by both the PBDR and IBDR1 programs.

4.2. Renewable energy community

The results discussed in this Section refer to the characteristic day of April. Fig. 14 displays the total costs and bills of the prosumers, without and with the incentive for the REC (Section 2.2.2), in all the couplings and operating conditions described in Section 4.1. Table 4 and Table 5 show the SC values of the REC and the economic benefits owing to the DR programs, without and with the incentive for the REC, respectively.

It emerges that Ind-Res is the best match in economic terms for the REC, except in the PBDR case in which the coupling Ter-Res leads to lower costs as shown in Fig. 14. In the REC with incentive, the total costs of the basic aggregation of Ind-Res are 722.65 €, while those of Ter-Res

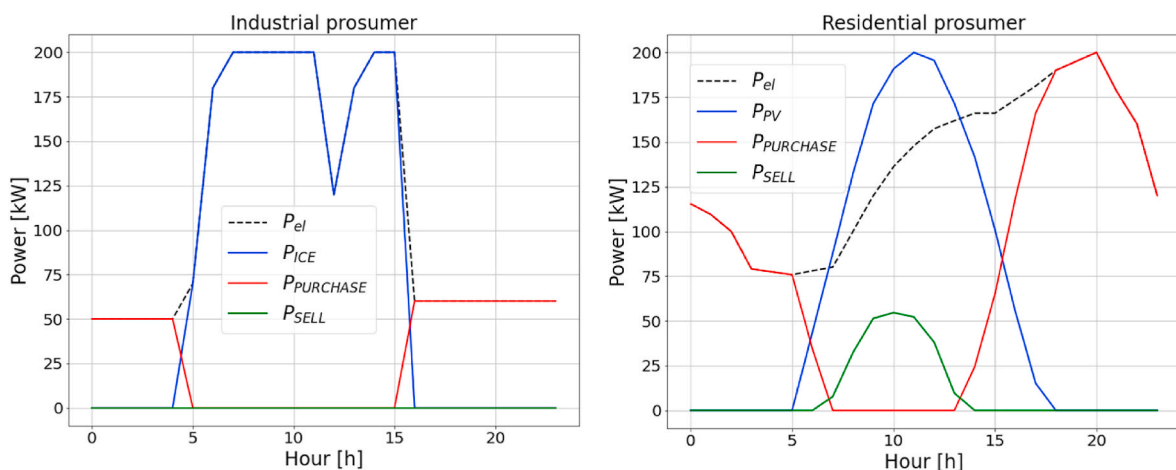


Fig. 10. Power flows in the separate operation of the industrial (left) and residential (right) prosumers.

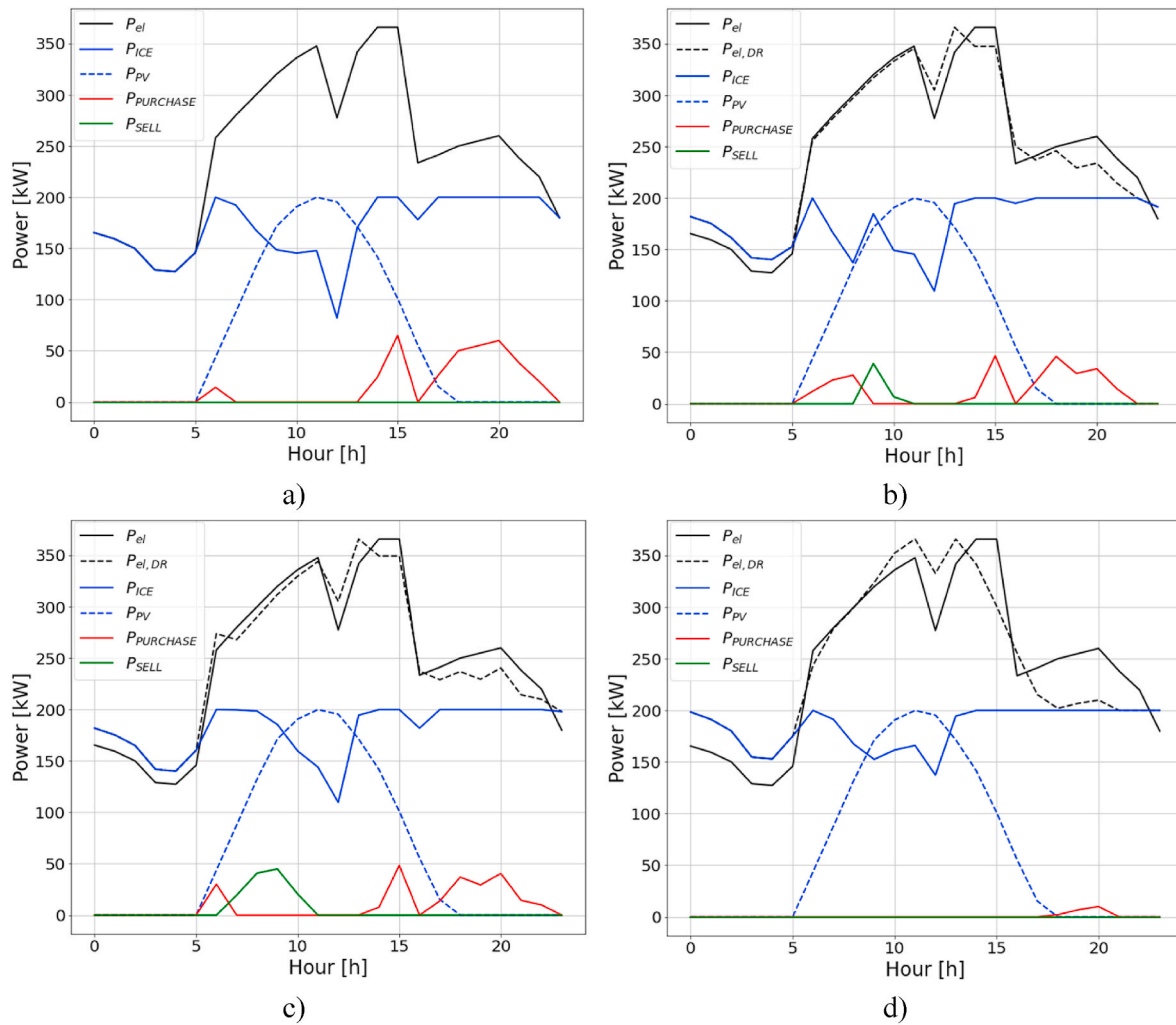


Fig. 11. Electrical demand, power generated and exchanged with the grid in the CEC for the Ind-Res coupling a) without DR, with b) PBDR, c) IBDR1 and d) IBDR2.

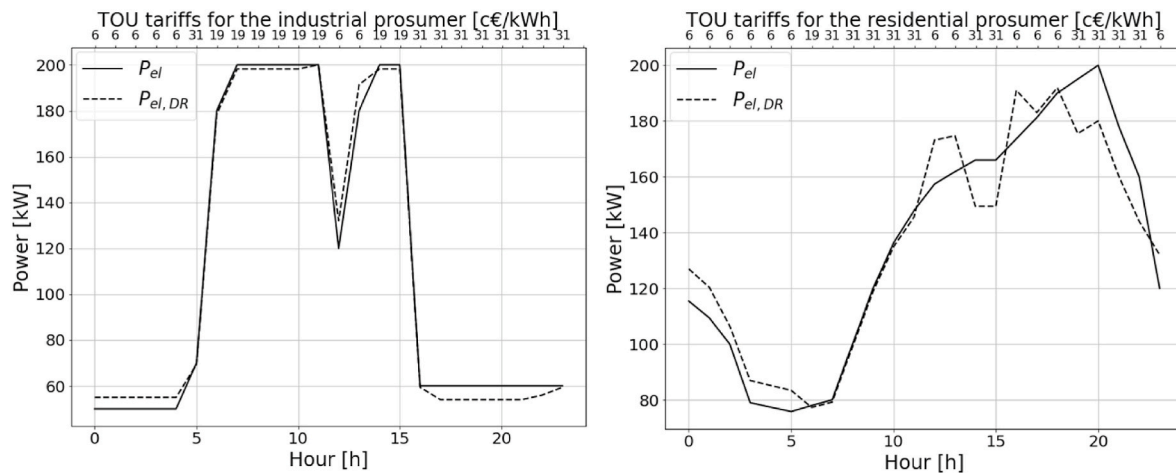


Fig. 12. Power demand before (continuous line) and after (dashed line) the application of the PBDR for the industrial (left) and residential (right) prosumers with the associated TOU tariffs.

and Agr-Res are 794.42 € and 896.02 €, respectively. Then, in the REC with incentive and implementation of the PBDR program, the total costs of Ter-Res (339.41 €) are 9% and 14% lower compared to those of Ind-Res (372.66 €) and Agr-Res (394.84 €), respectively. Indeed, in this case, the tertiary prosumer is more successful in profitably modifying its

demand profile, compared to the other prosumers, to obtain the lowest PBDR tariffs.

In the case of REC without incentive, enhancing the purchase of power from the grid is found to be more cost-effective than consuming biogas, resulting in the ICE exclusion from the optimal operation.

Table 4

Self-consumption and total economic savings, attained by the application of the DR programs compared to the basic aggregation, in the REC without incentive.

	SC (%)			Savings compared to basic aggregation (%)		
	Ind-Res	Agr-Res	Ter-Res	Ind-Res	Agr-Res	Ter-Res
Basic aggregation	25.32	21.83	24.18	–	–	–
PBDR	25.32	21.83	24.18	–47.77	–56.25	–52.15
IBDR1	25.32	21.83	24.18	–2.24	–2.26	–2.28
IBDR2	25.32	21.83	24.18	–4.24	–4.42	–4.23

Table 5

Self-consumption and total economic savings, attained by the application of the DR programs compared to the basic aggregation, in the REC with incentive.

	SC (%)			Savings compared to basic aggregation (%)		
	Ind-Res	Agr-Res	Ter-Res	Ind-Res	Agr-Res	Ter-Res
Basic aggregation	92.69	82.86	83.93	–	–	–
PBDR	88.9	76.8	81.09	–48.43	–55.93	–57.28
IBDR1	95.87	86.88	89.37	–3.51	–4.83	–4.8
IBDR2	99.05	89.35	92.47	–7.68	–8.65	–8.85

Consequently, a low and constant SC (from 21.83 to 25.32% across the different couplings) is obtained, as reported in Table 4, due to the constant and low generation from PV only (see Section 2.1), which is also much lower than the total daily energy demand. Moreover, the first criterion C1 benefits the residential prosumer only if the PBDR is applied or in the Agr-Res coupling, as shown in Fig. 14. For instance, considering the Agr-Res coupling in the REC without incentive and with the PBDR program, the bill of the residential prosumer (223.27 €) is 28% lower compared to that of the agricultural prosumer (308.73 €) when the cost allocation criterion C1 is applied. The application of the cost allocation criteria C1 and C2 to the REC with incentive leads to similar findings compared to the CEC, as reported in Figs. 14 and 9, respectively. For example, in the Ind-Res coupling with PBDR, the criterion C1 shows lower bills for the residential prosumer (125.88 € and 267.41 € for REC and CEC) compared to the industrial one (246.78 € and 366.16 € for REC and CEC).

Note that the SC of the REC with incentive (Table 5) decreases from the basic aggregation to the operating condition with PBDR (e.g., for Ter-Res the SC is reduced from 83.93% to 81.09%). Indeed, the TOU tariffs of the PBDR induce the hourly energy demand (which is also the energy purchased from the grid in the REC) to increase compared to the hourly energy generation. Fig. 13 shows that the difference between the hourly energy demand and generation is almost the same for the basic aggregation (red bars) and PBDR (green bars) in all the hours except for hour 15 where this difference is much higher in the PBDR. Consequently, the overall effect is a reduction of the SC compared to the basic aggregation, contrary to the CEC. Moreover, the IBDR1 and IBDR2 improve the SC (with incentive) and reduce the total costs (without and with incentive) of the community compared to the basic aggregation as reported in Table 5 and Fig. 14. In the REC with incentive, based on the Ter-Res coupling, applying the IBDR1 and IBDR2 programs increases the SC by 6% and 10% and reduces the total costs by 5% and 9%, respectively. Fig. 14 also shows that in the REC, both without and with incentive, the best DR program is the PBDR that approximately halves the total costs of the community respect to the case of basic aggregation without DR. For instance, in the REC with incentive, based on the Ter-Res coupling, the total costs noticeably decrease by 57% in the PBDR case respect to the basic aggregation.

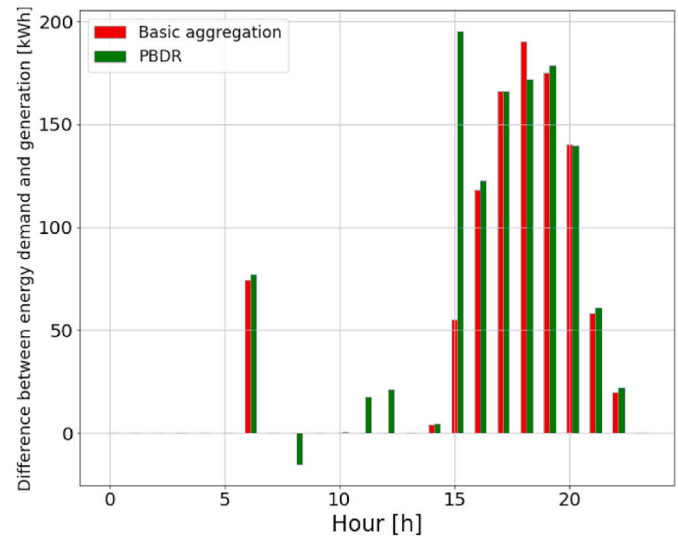


Fig. 13. Hourly difference between energy demand and generation in basic aggregation (red bars) and PBDR (green bars) in the REC with incentive for the Ter-Res coupling.

4.3. Critical remarks

The most relevant outcomes from the CEC and REC optimization, remarked hereinafter, are quite similar across the three characteristic days analysed (April, December, August).

In particular, the optimization results in the characteristic day of April show that:

- The CEC reports lower costs and higher self-consumption ratios (SC, Eq. (6)) compared to the REC in all couplings and operating conditions except for the PBDR case. In fact, the CEC purchases less power from the grid compared to the REC, leading to higher SC values. For example, in the basic aggregation (i.e., without DR) with Ind-Res coupling, the CEC (SC of 94.07%) has total costs 12–38% lower than the REC without incentive (SC of 92.69–25.32%), respectively. Conversely, with the PBDR, the REC leads to lower costs than the CEC since it benefits from advantageous DR tariffs for its total power demand, which corresponds to the overall power purchased from the grid (contrary to the CEC). Moreover, the CEC configuration presents the highest value of SC equal to 99.68%, which is obtained with the coupling Ind-Res and with application of the IBDR2 program.
- In almost all cases (i.e., CEC and REC configurations with or without DR programs), Ind-Res is found to be the prosumer coupling leading to the lowest costs. For example, in the CEC the total costs of the basic aggregation of Ind-Res are 21% and 11% lower than those of Agr-Res and Ter-Res, respectively.
- The cost allocation criterion C1 successfully leads to lower bills for the residential prosumer (compared to the other prosumer in each coupling) in both CEC and REC with incentive, whereas in the REC without incentive only in the coupling Agr-Res or when the PBDR is applied. Considering the basic aggregation of CEC (or REC with incentive), the application of the novel criterion C1 to the best coupling Ind-Res and criterion C2 to the worst coupling Agr-Res results in a bill of the residential prosumer equipped with PV that is 28% (or 17% for REC with incentive) and 10% lower than that of the industrial and agricultural one equipped with ICE, respectively.
- In the CEC, the PBDR and IBDR1 programs result to reduce the total costs of 0.6 and 2.7% (Ind-Res and Agr-Res), whereas the IBDR2 program of approximately 4%. In the REC (without and with incentive), the PBDR results to be the most effective program with cost savings of 48 and 56% (Ind-Res and Agr-Res), whereas the IBDR programs do not allow obtaining savings higher than 8 and 9% (Ind-Res and Agr-Res).

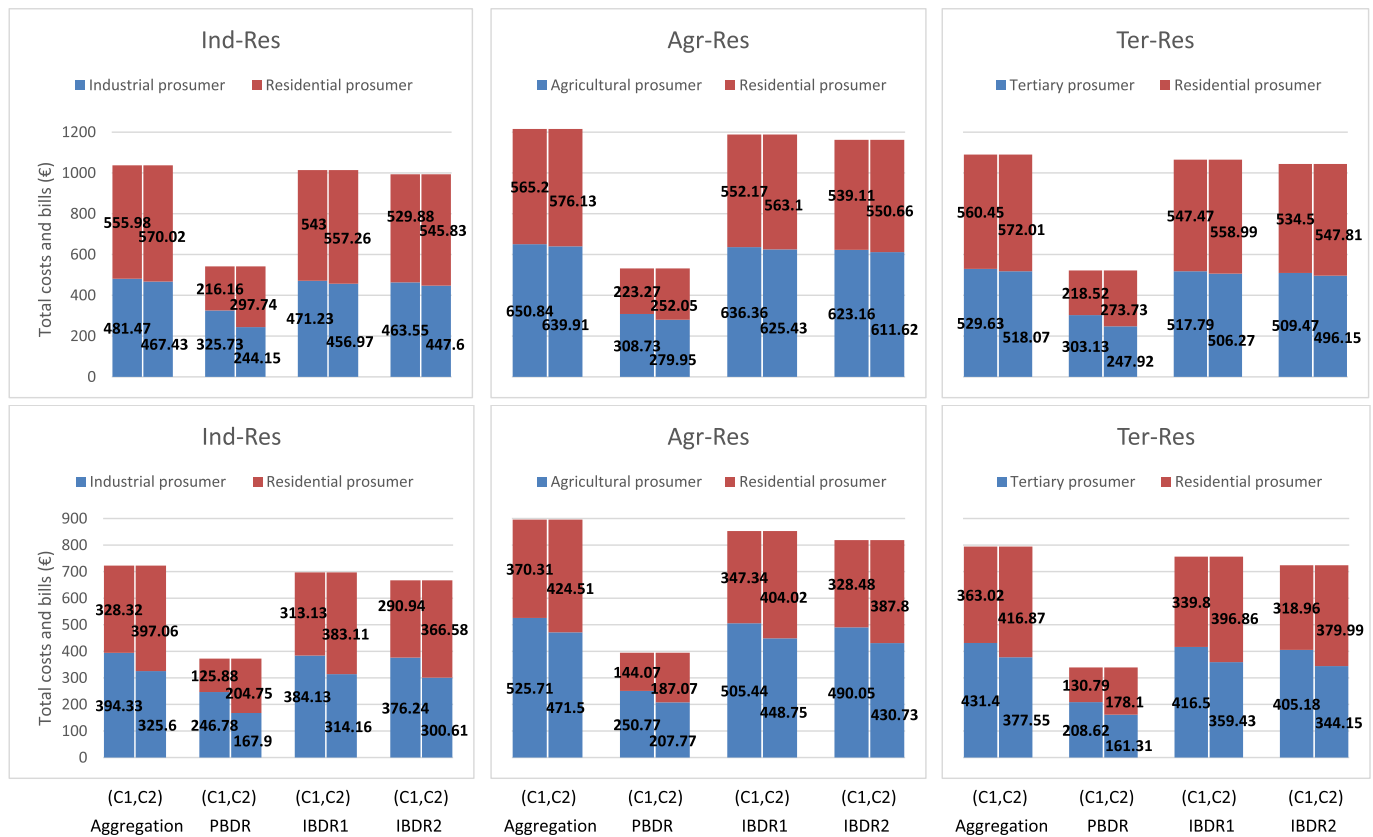


Fig. 14. Economic results from the REC optimization without (above) and with incentive (below) in April.

Moreover, in the CEC the DR programs always improve the complementarity between prosumers and, in turn, the SC (by 2–6% and 5–8% in Ind-Res and Agr-Res). On the other hand, in the REC with incentive, only the IBDR programs enhance the SC with respect to the basic aggregation (by 3–7% and 5–8% in Ind-Res and Agr-Res) contrary to the PBDR (–4% and –7% in Ind-Res and Agr-Res). In fact, the economically convenient PBDR tariffs promote the purchase of a higher amount of power from the grid, therefore lowering the SC of the REC.

On the other hand, for both the characteristic days of December and August, Figures A.1–A.2–A.3 in the Appendix show the total costs of the community and the bills of the prosumers in the CEC, REC without and REC with incentive, respectively. The main findings obtained are consistent with those referring to the characteristic day of April. In the basic aggregation, the total costs of the CEC are 14–29% and 12–39% lower than those of the REC with/without incentive in December and August, respectively. In December the coupling Ind-Res shows 17, 14, 16% (19, 15, 19% in August) lower costs compared to Agr-Res in the basic aggregation of CEC, REC without and REC with incentive, respectively. In December, owing to the low solar irradiance, the proposed cost allocation criterion does not benefit the residential prosumer equipped with PV, whereas in August its bill is 20% (REC with incentive) and 30% (CEC) lower than that of the industrial prosumer. Again, the IBDR2 and PBDR are the most effective programs in the CEC and REC, respectively, with maximum economic savings of 2.3–4.7% (December–August) and 61.8–57.8% (December–August). Eventually, note that the total costs of CEC and REC with incentive significantly reduce from the case of December to April, due to the increase of the solar irradiance and of the hourly selling prices (see Fig. 3). Conversely, these total costs slightly decrease from April to August because the decline of the selling prices (Fig. 3) counteracts the effect of rise of the solar irradiance.

5. Conclusions

This paper focuses on the optimal economic aggregation of prosumers within the Citizen and Renewable Energy Communities (CEC and REC, respectively). The innovative feature of this work, which widens the main perspective of the current literature, relies on the identification and comprehensive assessment of three main aspects affecting the economic convenience of Energy Communities (ECs): the complementarity between energy demand and generation profiles of prosumers, the criterion to allocate the operating costs of the EC among its members and the application of Demand Response (DR) programs. Differently from the main literature, this paper assesses these aspects together by an optimization procedure. The main findings of this work allow pointing out the following general guidelines to achieve an optimal aggregation of prosumers within an EC, resulting in economic savings for both the community and its single members:

1. A good complementarity between different prosumers is crucial to exploit conveniently the combination of dispatchable and not-dispatchable energy sources. For instance, coupling a residential prosumer equipped with PV and another prosumer among the industrial-agricultural-tertiary ones equipped with an ICE, allows the prosumer with PV consuming dispatchable power from ICE when solar energy is not available, while covering the potential demand peaks of the prosumer with ICE by PV. *Ind-Res and Agr-Res are the demand couplings featured by the higher and lower complementarity, respectively, with the former generating costs 17–21%, 14–15%, 16–19% (across the three characteristic days) lower than the latter for basic aggregations of CEC, REC without and REC with incentive.*
2. In addition to complementarity, a fair and effective cost allocation criterion among the EC members must be applied. Considering the characteristic days of April and August, the novel criterion C1 results to be more successful in assigning lower costs to the more virtuous

- prosumer who exploits a free-of-charge and not dispatchable renewable source (i.e., solar) compared to the traditional criterion C2, which allocates higher costs to prosumers having higher energy demands. *The implementation of the cost allocation criterion C1 permits the residential prosumer equipped with PV to benefit from a bill that is 17–20% (April–August), in the REC with incentive, and 28–30% (April–August), in the CEC, lower compared to that of the industrial prosumer.*
3. In addition to the complementarity and independently of the cost allocation criterion, further economic savings are obtained at the community scale by profitable modifications of the demand profiles with DR programs, for given sources and design of the energy conversion units. *The IBDR2 and PBDR are the best DR options for the CEC and REC, with economic savings of 4–5% and 48–62% (across the three characteristic days) compared to the basic aggregation, respectively.*

In summary, the general guidelines show that: 1) *complementarity may reduce costs in the order of 15–20%, 2) a fairer cost allocation criterion among EC members may reduce the bills of prosumers using free-of-charge renewables by 20–30% compared to those using dispatchable sources, and 3) price-based DR programs may reduce EC costs even beyond 50%.* These numbers supply a realistic quantitative indication of the economic advantages deriving from building an EC, and of the need to consider all the mentioned aspects together for the correct evaluation of its positive economic impact. Note that the combinations of prosumers, having demand profiles with characteristic shapes related to different sectors, within different configurations of ECs and the operation optimization carried out in typical seasonal days concurrently guarantee the general validity of the achieved guidelines. Moreover, the general guidelines are formulated on the strength of the global solutions attained by the optimization procedure, which is solved by the linear-programming based branch-and-bound algorithm ensuring the global optimality of the solutions.

Eventually, the CEC is the best configuration in terms of lowest total costs (12–39% lower than those of the REC) as well as highest self-consumption values (till 99.68%). On the other hand, the REC (supported by a government incentive) is economically preferred to the CEC when the PBDR is applied. From the obtained general guidelines, policy makers could develop energy policies to propose new market mechanisms, including tailored incentives and DR programs, with the aim of

fostering the spread of the RECs and enhancing their economic competitiveness with respect to the CECs.

The obtained guidelines could pave the way for the development of a wider modelling tool able to assess the impact of a large diffusion of local ECs on a national/European power system, identifying the role of the ECs in providing flexibility to the power system. This modelling tool could also allow addressing the economic evaluation of larger aggregations of energy users compared to those studied in this work, still ensuring acceptable computational times to solve the optimization procedure. Further research could also deal with the optimization of the EC configurations under uncertainty (e.g., that of intermittent renewables) and the implementation of other innovative cost allocation criteria.

Credit author statement

Gabriele Volpato: Software, Writing – original draft, Methodology, Conceptualization, Writing – review & editing; Gianluca Carraro: Conceptualization, Supervision, Methodology, Formal analysis, Writing – original draft; Marco Cont: Software, Formal analysis; Piero Danieli: Visualization, Writing – review & editing; Sergio Rech: Visualization, Writing – review & editing; Andrea Lazzaretto: Conceptualization, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Appendix

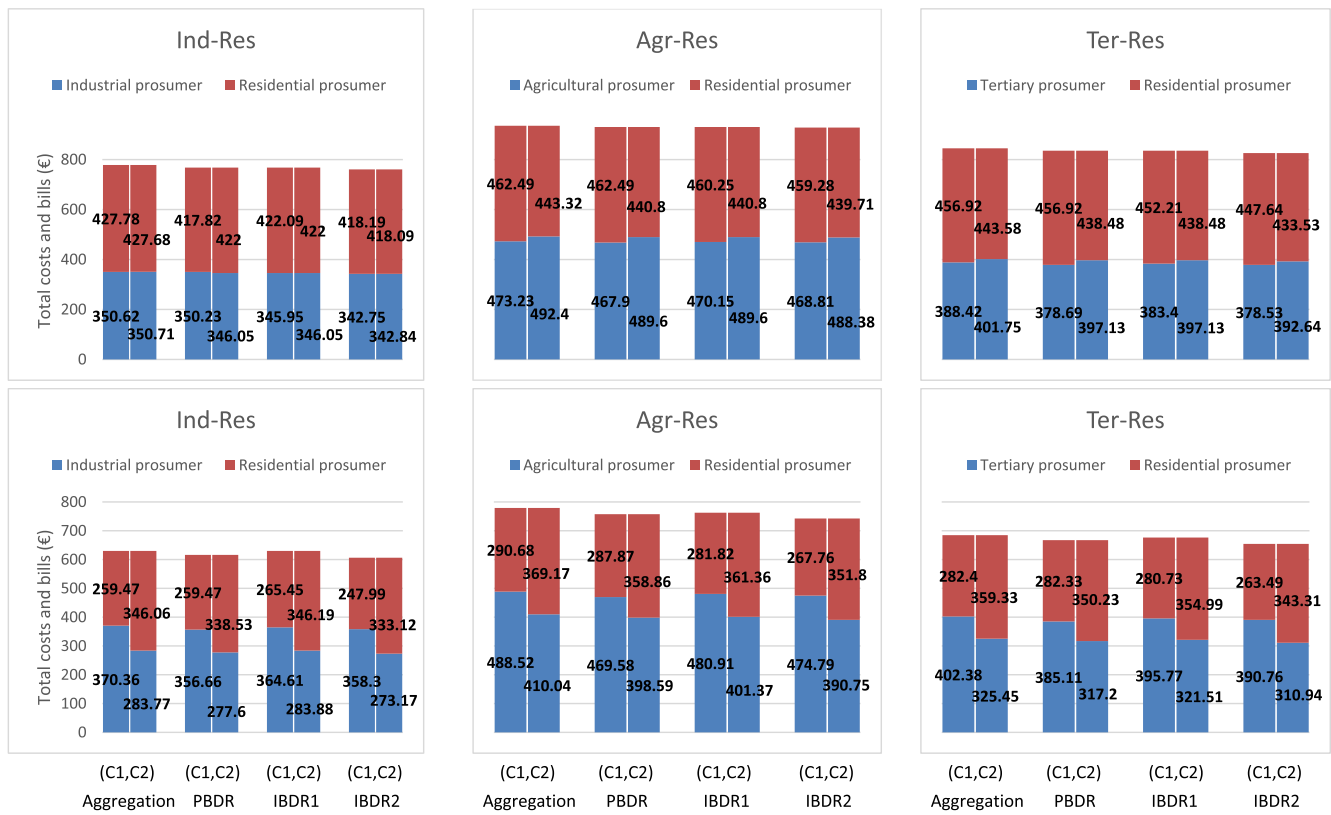


Fig. A.1. Economic results from the CEC optimization in December (above) and August (below).

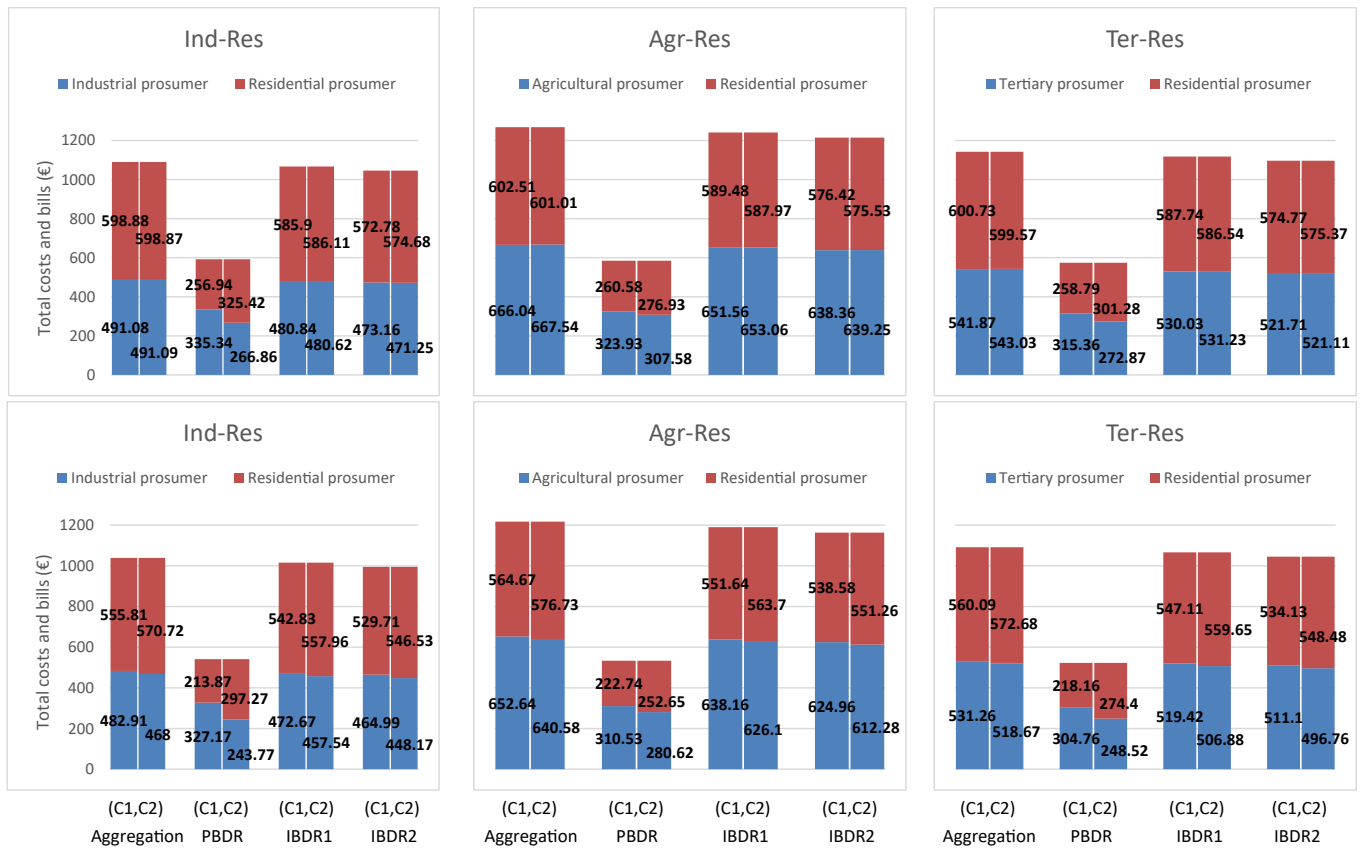


Fig. A.2. Economic results from the optimization of REC without incentive in December (above) and August (below).

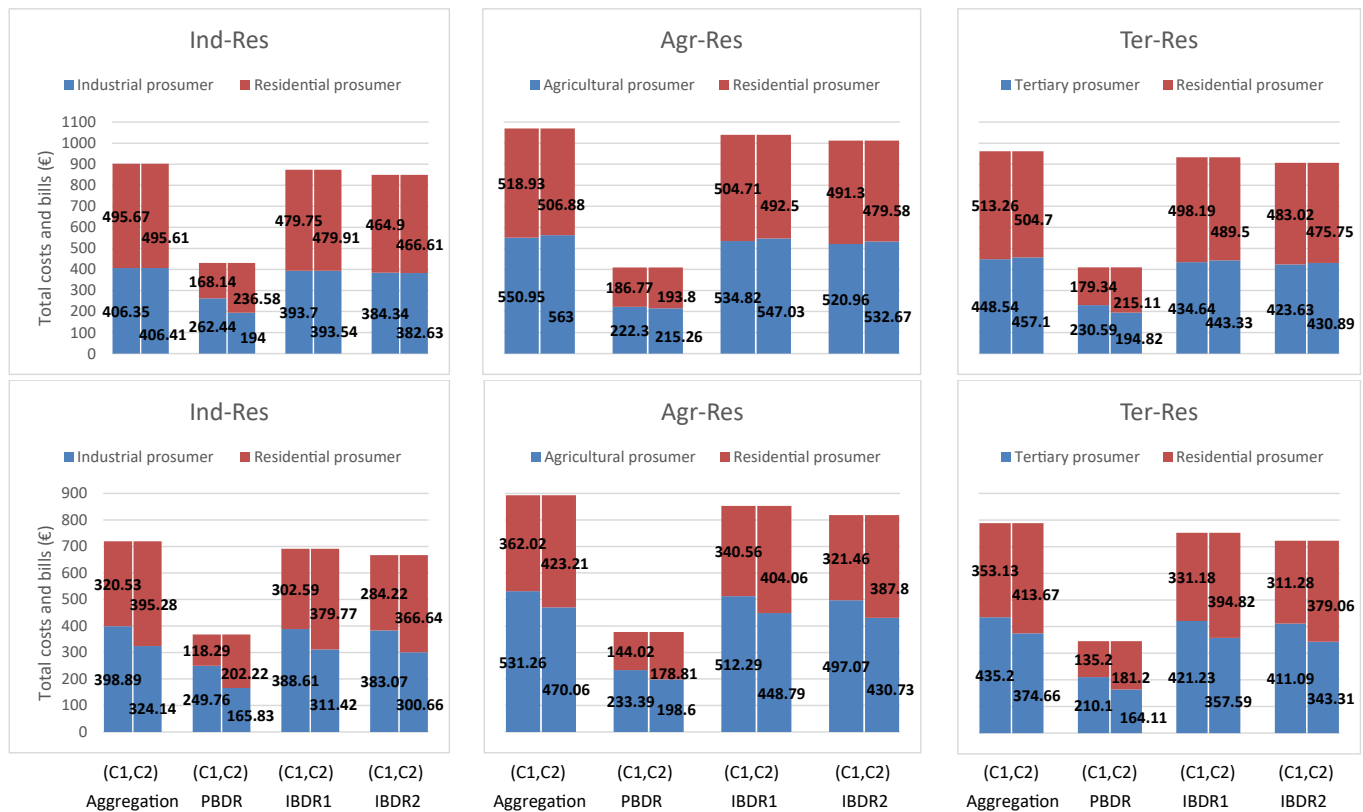


Fig. A.3. Economic results from the optimization of REC with incentive in December (above) and August (below).

References

- [1] European commission clean energy for all Europeans package. 2019 https://energy.ec.europa.eu/topics/energy-strategy/clean-energy-all-europeans-package_en. [Accessed 11 May 2022].
- [2] European commission's communication COM 2014. A Policy Framework for Climate and Energy in the Period from 2020 to 2030. 2014. <https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=celex%3A52014DC0015>. [Accessed 11 May 2022].
- [3] IEA. World energy outlook 2020. <https://www.iea.org/reports/world-energy-outlook-2020>. [Accessed 11 May 2022].
- [4] European Commission's Communication COM(2021) 550 final. "Fit for 55": delivering the EU's 2030 Climate Target on the way to climate neutrality. 2021 <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52021DC0550>. [Accessed 11 May 2022].
- [5] European commission's communication COM/2020/562 final. Stepping up Europe's 2030 climate ambition investing in a climate-neutral future for the benefit of our people. 2020. <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52020DC0562>. [Accessed 11 May 2022].
- [6] De Crescenzo V, Baratta R, Simeoni F. Citizens' engagement in funding renewable and energy efficiency projects: a fuzzy set analysis. *J Clean Prod* 2020;277:124060.
- [7] Directive (EU) 2019/944 of the European Parliament and of the Council of 5 June 2019 on Common Rules for the Internal Market for Electricity and Amending Directive 2012/27/EU (Recast). 2019 <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32019L0944>. [Accessed 11 May 2022].
- [8] Directive (EU) 2018/2001 of the European parliament and of the council of 11 december 2018 on the promotion of the use of energy from renewable sources (recast). 2018 https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=uriserv:OJ.L_.2018.328.01.0082.01.ENG. [Accessed 11 May 2022].
- [9] Reis IF, Gonçalves I, Lopes MA, Antunes CH. A multi-agent system approach to exploit demand-side flexibility in an energy community. *Util Pol* 2020;67:101114.
- [10] Caramizaru E, Uihlein A. Energy communities: an overview of energy and social innovation. Luxembourg: EUR 30083 EN, Publications Office of the European Union; 2020:JRC119433. doi:10.2760/180576 (online).
- [11] Candelise C, Ruggieri G. Status and evolution of the community energy sector in Italy. *Energies* 2020;13(8):1888.
- [12] de Brauwier CP, Cohen J. Analysing the potential of citizen-financed community renewable energy to drive Europe's low-carbon energy transition. *Renew Sustain Energy Rev* 2020;133:110300.
- [13] TESTO COORDINATO DEL DECRETO-LEGGE 30 dicembre 2019 n 162 art 42-bis. GSE; 2020. <https://www.gse.it/servizi-per-te/autoconsumo/gruppi-di-autoconsumatori-e-comunita-di-energia-rinnovabile/documenti>. [Accessed 11 May 2022].
- [14] Lowitzsch J, Hoicka C, Van Tulder F. Renewable energy communities under the 2019 European Clean Energy Package—Governance model for the energy clusters of the future? *Renew Sustain Energy Rev* 2020;122:109489.
- [15] Guo J, Zhang P, Wu D, Liu Z, Liu X, Zhang S, Yang X, Ge H. Multi-objective optimization design and multi-attribute decision-making method of a distributed energy system based on nearly zero-energy community load forecasting. *Energy* 2022;239:122124.
- [16] Fina B, Auer H, Friedl W. Profitability of PV sharing in energy communities: use cases for different settlement patterns. *Energy* 2019;189:116148.
- [17] Fleischhacker A, Lettner G, Schwabeneder D, Auer H. Portfolio optimization of energy communities to meet reductions in costs and emissions. *Energy* 2019;173:1092–105.
- [18] Islam M, Das BK, Das P, Rahaman MH. Techno-economic optimization of a zero emission energy system for a coastal community in Newfoundland, Canada. *Energy* 2021;220. 119709.
- [19] Sokolnikova P, Lombardi P, Arendarski B, Suslov K, Pantaleo A, Kranhold M, Komarnicki P. Net-zero multi-energy systems for Siberian rural communities: a methodology to size thermal and electric storage units. *Renew Energy* 2020;155: 979–89.
- [20] Kim M-H, Kim D, Heo J, Lee D-W. Techno-economic analysis of hybrid renewable energy system with solar district heating for net zero energy community. *Energy* 2019;187:115916.
- [21] Rossi F, Heleno M, Basosi R, Sinicropi A. LCA driven solar compensation mechanism for Renewable Energy Communities: the Italian case. *Energy* 2021;235: 121374.
- [22] Balderrama S, Lombardi F, Stevanato N, Peña G, Colombo E, Quoilin S. Surrogate models for rural energy planning: application to Bolivian lowlands isolated communities. *Energy* 2021;121108.
- [23] Hanke F, Lowitzsch J. Empowering vulnerable consumers to join renewable energy communities—towards an inclusive design of the clean energy package. *Energies* 2020;13(7):1615.
- [24] Di Lorenzo G, Rotondo S, Araneo R, Petrone G, Martirano L. Innovative power-sharing model for buildings and energy communities. *Renew Energy* 2021;172: 1087–102.
- [25] Fichera A, Marrasso E, Sasso M, Volpe R. Energy, environmental and economic performance of an urban community hybrid distributed energy system. *Energies* 2020;13(10):2545.

- [26] Ceglia F, Marrasso E, Roselli C, Sasso M. Small renewable energy community: the role of energy and environmental indicators for power grid. *Sustainability* 2021;13(4):2137.
- [27] Bartolini A, Carducci F, Muñoz CB, Comodi G. Energy storage and multi energy systems in local energy communities with high renewable energy penetration. *Renew Energy* 2020;159:595–609.
- [28] Ugwoke B, Corgnati S, Leone P, Borchiellini R, Pearce J. Low emissions analysis platform model for renewable energy: community-scale case studies in Nigeria. *Sustain Cities Soc* 2021;67:102750.
- [29] Barone G, Buonomano A, Forzano C, Giuzio GF, Palombo A. Increasing renewable energy penetration and energy independence of island communities: a novel dynamic simulation approach for energy, economic, and environmental analysis, and optimization. *J Clean Prod* 2021;127558.
- [30] Liu Z, Fan G, Sun D, Wu D, Guo J, Zhang S, Yang X, Lin X, Ai L. A novel distributed energy system combining hybrid energy storage and a multi-objective optimization method for nearly zero-energy communities and buildings. *Energy* 2021;122577.
- [31] Kotagodahetti R, Hewage K, Karunathilake H, Sadiq R. Evaluating carbon capturing strategies for emissions reduction in community energy systems: a life cycle thinking approach. *Energy* 2021;232(C).
- [32] Li J, Liu P, Li Z. Optimal design and techno-economic analysis of a solar-wind-biomass off-grid hybrid power system for remote rural electrification: a case study of west China. *Energy* 2020;208:118387.
- [33] Aviso KB, Marfori III IAV, Tan RR, Ubando AT. Optimizing abnormal operations of off-grid community utility systems with fuzzy P-graph. *Energy* 2020;202:117725.
- [34] Lund PD, Lindgren J, Mikkola J, Salpakari J. Review of energy system flexibility measures to enable high levels of variable renewable electricity. *Renew Sustain Energy Rev* 2015;45:785–807.
- [35] Niu J, Tian Z, Zhu J, Yue L. Implementation of a price-driven demand response in a distributed energy system with multi-energy flexibility measures. *Energy Convers Manag* 2020;208:112575.
- [36] Guo Z, Ji Z, Wang Q. Blockchain-enabled demand response scheme with individualized incentive pricing mode. *Energies* 2020;13(19):5213.
- [37] Jordehi AR. Optimisation of demand response in electric power systems, a review. *Renew Sustain Energy Rev* 2019;103:308–19.
- [38] El Geneidy R, Howard B. Contracted energy flexibility characteristics of communities: analysis of a control strategy for demand response. *Appl Energy* 2020;263:114600.
- [39] Qiu R, Liao Q, Yan J, Yan Y, Guo Z, Liang Y, Zhang H. The coupling impact of subsystem interconnection and demand response on the distributed energy systems: a case study of the composite community in China. *Energy* 2021;228:120588.
- [40] Berka AL, Creamer E. Taking stock of the local impacts of community owned renewable energy: a review and research agenda. *Renew Sustain Energy Rev* 2018; 82:3400–19.
- [41] Gjorgievski VZ, Cundeva S, Georgiou GE. Social arrangements, technical designs and impacts of energy communities: a review. *Renewable Energy*; 2021.
- [42] Li N, Hakvoort RA, Lukszo Z. Cost allocation in integrated community energy systems-A review. *Renew Sustain Energy Rev* 2021;144:111001.
- [43] Wang Y, Qi C, Dong H, Wang S, Wang X, Zeng M, Zhu J. Optimal design of integrated energy system considering different battery operation strategy. *Energy* 2020;212:118537.
- [44] Li P, Wang Z, Wang J, Yang W, Guo T, Yin Y. Two-stage optimal operation of integrated energy system considering multiple uncertainties and integrated demand response. *Energy* 2021;225:120256.
- [45] Li P, Wang Z, Liu H, Wang J, Guo T, Yin Y. Bi-level optimal configuration strategy of community integrated energy system with coordinated planning and operation. *Energy* 2021;236:121539.
- [46] Cosic A, Stadler M, Mansoor M, Zellinger M. Mixed-integer linear programming based optimization strategies for renewable energy communities. *Energy* 2021; 237:121559.
- [47] Jiang A, Yuan H, Li D. Energy management for a community-level integrated energy system with photovoltaic prosumers based on bargaining theory. *Energy* 2021;225:120272.
- [48] Rech S. Smart energy systems: guidelines for modelling and optimizing a fleet of units of different configurations. *Energies* 2019;12(7):1320.
- [49] Gurobi optimization. <https://www.gurobi.com/>. [Accessed 11 May 2022].
- [50] Gabrielli P, Gazzani M, Martelli E, Mazzotti M. Optimal design of multi-energy systems with seasonal storage. *Appl Energy* 2018;219:408–24.
- [51] IEA, International Energy Agency - IEA. Residential prosumers - drivers and policy options (re-prosumers). IEA; 2014. http://iea-retd.org/wp-content/uploads/2014/06/RE-PROSUMERS IEA-RETD_2014.pdf. [Accessed 11 May 2022].
- [52] Celli G, Mocci S, Pilo F, Soma GG. A multi-objective approach for the optimal distributed generation allocation with environmental constraints. In: Proceedings of the 10th international conference on probabilistic methods applied to power systems. IEEE; 2008. p. 1–8.
- [53] Generac industrial power. <https://www.generac.com/Industrial/products/gaseous-s-generators/configured?filter=5>. [Accessed 11 May 2022].
- [54] Photovoltaic geographical information system (PVGIS). <https://ec.europa.eu/jrc/en/pvgis>. [Accessed 11 May 2022].
- [55] ARERA, Autorità di Regolazione per Energia Reti e Ambiente. <https://www.arera.it/it/inglese/index.htm>. [Accessed 11 May 2022].
- [56] Mercati Energetici Gestore. <https://www.mercatoelettrico.org/en/default.aspx>. [Accessed 11 May 2022].
- [57] van Bommel N, Höffken JI. Energy justice within, between and beyond European community energy initiatives: a review. *Energy Res Social Sci* 2021;79:102157.
- [58] Monfared HJ, Ghasemi A, Loni A, Marzband M. A hybrid price-based demand response program for the residential micro-grid. *Energy* 2019;185:274–85.
- [59] Kohlhepp P, Harb H, Wolisz H, Waczowicz S, Müller D, Hagenmeyer V. Large-scale grid integration of residential thermal energy storages as demand-side flexibility resource: a review of international field studies. *Renew Sustain Energy Rev* 2019; 101:527–47.
- [60] Luo Z, Hong S, Ding Y. A data mining-driven incentive-based demand response scheme for a virtual power plant. *Appl Energy* 2019;239:549–59.
- [61] Zhan Y, Zheng QP, Wang J, Pinson P. Generation expansion planning with large amounts of wind power via decision-dependent stochastic programming. *IEEE Trans Power Syst* 2016;32(4):3015–26.
- [62] Shewale A, Mokhade A, Funde N, Bokde ND. An overview of demand response in smart grid and optimization techniques for efficient residential appliance scheduling problem. *Energies* 2020;13(16):4266.
- [63] Pazouki S, Haghifam M-R, Moser A. Uncertainty modeling in optimal operation of energy hub in presence of wind, storage and demand response. *Int J Electr Power Energy Syst* 2014;61:335–45.
- [64] Nosratabadi SM, Hooshmand R-A, Gholipour E. A comprehensive review on microgrid and virtual power plant concepts employed for distributed energy resources scheduling in power systems. *Renew Sustain Energy Rev* 2017;67: 341–63.
- [65] Harsh P, Das D. Energy management in microgrid using incentive-based demand response and reconfigured network considering uncertainties in renewable energy sources. *Sustain Energy Technol Assessments* 2021;46:101225.